Deep Learning for Neurodegenerative Disease detection through Spontaneous Speech

Master Thesis

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Abstract

Speech is an attractive non-invasive biomarker for neurodegenerative disease, but multimodal systems that combine acoustic and textual signals expose many interacting design choices whose individual contributions are rarely isolated. This thesis treats such a system as a composition of largely independent design axes (modality, fusion, audio–text alignment, Wav2Vec layer pooling, layer mixing, and feature extractor) and estimates the contribution of each through controlled ablations across three clinical tasks: Alzheimer's disease classification on ADReSSo, amyloid-β status prediction, and primary progressive aphasia variant classification. The two Spanish tasks are defined on a spontaneous-speech subset of the established SPIN clinical cohort, curated for this work with confirmed amyloid status and variant labels and used here for the first time, which the thesis contributes as a new clinical resource. Its central methodological contribution is a pair of token-to-token alignment schemes, uniform and proportional, that refine the word-to-token alignment of prior work at the subword level. The strongest per-axis choices are consolidated into a single configuration-driven implementation, yielding a recommended architecture that is competitive across tasks while remaining lightweight and deployable.

1 Introduction

Alzheimer's disease (AD) is the most common cause of dementia worldwide. Its underlying pathology, the progressive build-up of amyloid-β plaques and neurofibrillary tau tangles, begins years before any clinical symptom appears [1], which is why detecting it early has become a central goal of current research. Confirmation is the bottleneck: a definitive read on amyloid status still depends on a lumbar puncture or a positron emission tomography (PET) scan, neither of which scales to routine screening. Diagnosis at the syndrome level is no easier. Primary progressive aphasia (PPA) comes in three canonical variants, non-fluent/agrammatic, semantic, and logopenic [2, 3], whose behavioural profiles overlap enough that telling them apart is difficult, especially in early or atypical cases [4]. Scalable amyloid screening and reliable PPA subtyping thus point to the same need: non-invasive computational tools that support, rather than replace, expert judgement.

Speech is a natural place to look. Producing a spontaneous description draws at once on cognitive planning, lexical access, syntactic construction, motor execution, and prosodic control, a chain of processes that neurodegeneration disrupts unevenly and so leaves its traces in the signal. Those traces are split across two channels that a single system can read at once: the fine temporal micro-structure of speech (planning pauses, articulatory slowing, reduced informational density) and its lexical–semantic content [5], a division that already motivates treating the problem as multimodal. And unlike a scan or an assay, a recording costs almost nothing to collect and can be gathered repeatedly, even remotely [6]. The clinical infrastructure is already in place: structured prompts such as the Picnic Scene from the Western Aphasia Battery [7] are routine in neuropsychological batteries, and the recordings they yield can be analysed computationally with no extra instrumentation.

The first computational attempts leaned on handcrafted linguistic descriptors and a few shallow acoustic statistics; they proved the idea workable but generalised poorly [8]. Deep learning raised the ceiling. Self-supervised encoders, Wav2Vec 2.0 on the acoustic side [9] and BERT-family models on the textual [10], supply far richer representations than any hand-built feature set. Shared benchmarks made the gains measurable: the ADReSSo challenge [11] fixed splits and protocols for like-for-like comparison, and the surveys built on this line of work converge on one finding in particular, that combining acoustic and textual evidence beats either modality alone [12].

Yet a headline accuracy number hides as much as it reveals. A typical pipeline bundles several decisions at once, which modalities to use, how to fuse them, how to align acoustic frames with text tokens, how to configure the self-supervised encoder, and then reports a single end-to-end figure. With the choices entangled inside one system, there is no way to tell which technique earned the performance and which merely came along for the ride on a particular dataset or implementation. A second blind spot is linguistic. Most benchmarked systems are built and tested on English, and whether their trends survive in languages with different morphology and prosody is largely unknown. Pathology-confirmed cohorts in other languages do exist, among them the Spanish-speaking Sant Pau Initiative on Neurodegeneration (SPIN) [13], but they have drawn little attention in the multimodal deep-learning literature.

1.1 Motivation and objectives

These two gaps, entangled design choices and an English-centric evidence base, set the agenda for this thesis. The premise is straightforward: if published systems differ along every axis at once, the only way to learn what each axis contributes is to move one at a time. A structured ablation programme, varying a single design choice while the rest stay pinned to controlled defaults, turns a tangle of confounded comparisons into per-axis evidence that can guide later design.

An earlier companion study [14] reported first results for AD and PPA classification on the ADReSSo and SPIN cohorts; the present thesis builds on it by breaking the pipeline into design choices that can each be ablated in isolation. That paper was published partway through this work, before the full ablation programme was complete, so some of its figures differ from the final results reported here, which supersede them. The thesis's main objectives are:

1. **Clinical speech resource.** Curate and characterise a Spanish-speaking spontaneous-speech subset of the established SPIN cohort [13], paired with confirmed amyloid-β status and PPA variant labels and used here for the first time, to complement the public English ADReSSo benchmark.

2. **Modality.** Quantify the individual and combined contribution of audio, text, and handcrafted acoustic features across three clinical speech tasks: AD classification (ADReSSo, English), amyloid-β status prediction (SPIN, Spanish, validation-only), and PPA variant classification (the same SPIN cohort, WAB picture-description subset, Spanish, validation-only).

3. **Fusion.** Systematically compare seven sequence-to-sequence fusion families (self-attention, multi-head self-attention, transformer with positional encoding, cross-attention, gated cross-attention, bidirectional cross-attention, and a Mamba state-space comparator), on all three tasks, taking bidirectional cross-attention (the fusion introduced by CogniAlign) as the grid's default and replicating the trends under a second, independent implementation.

4. **Alignment.** Propose and evaluate two novel token-to-token audio–text alignment schemes (uniform and proportional) that operate at the subword level, benchmarked against the word-to-token alignment of CogniAlign [15]; this is the central methodological contribution of the thesis.

5. **Wav2Vec pooling and layer mixing.** Evaluate final-layer versus weighted multi-layer Wav2Vec pooling, and the adaptive per-utterance *sample* layer mixer that refines it, evaluated in the unified implementation.

6. **Feature extractor.** Conduct a single-factor encoder ablation that swaps one audio or text backbone at a time while holding the rest of the pipeline fixed, spanning 24 runs (four encoder variants $\times$ three tasks $\times$ two seeds).

7. **Recommended architecture.** Synthesise the strongest per-axis choices into a recommended hybrid architecture, drawn together in the conclusions.

The programme works through the axes in stages, in turn a modality, fusion, alignment, layer-pooling, and combined alignment–pooling study, each set out in full in the methodology (Section 3.7). The strongest choice from each stage is then assembled and re-validated in a single configuration-driven implementation, `res_model`. A second stack, the Double Multi-Head Attention Multimodal (DMHAM), plays only a supporting role: it has no alignment axis and is not a rival to be beaten, but an independent codebase on which the fusion and pooling trends can be checked for reproduction. What the thesis compares throughout is the technique, not the software that happens to implement it.

The remainder of this thesis is organised as follows. Section 2 reviews the state of the art at the intersection of neurodegenerative-disease biomarkers, clinical linguistics, and multimodal deep learning on spontaneous speech. Section 3 formalises the prediction problem as a composition of six design axes, describes the three clinical cohorts and their preprocessing, the system architecture and its three implementations, the training protocol, and the evaluation metrics. Section 4 reports the staged ablation programme and the unified-model validation, and Section 5 synthesises the findings into the recommended architecture and outlines future work.

1.2 Work plan

The project ran from February to June 2026 in the broadly sequential phases shown in Figure 1. It began with a literature review and the data preprocessing and embedding-cache setup, then built the two primitive stacks (CogniAlign and DMHAM) in which the individual techniques were first developed. The per-axis ablations followed, first fusion, pooling, and alignment, then the layer-mixer and encoder studies, after which results analysis consolidated the findings into the recommended configuration. Thesis writing and defence preparation run alongside and after the later experimental work.

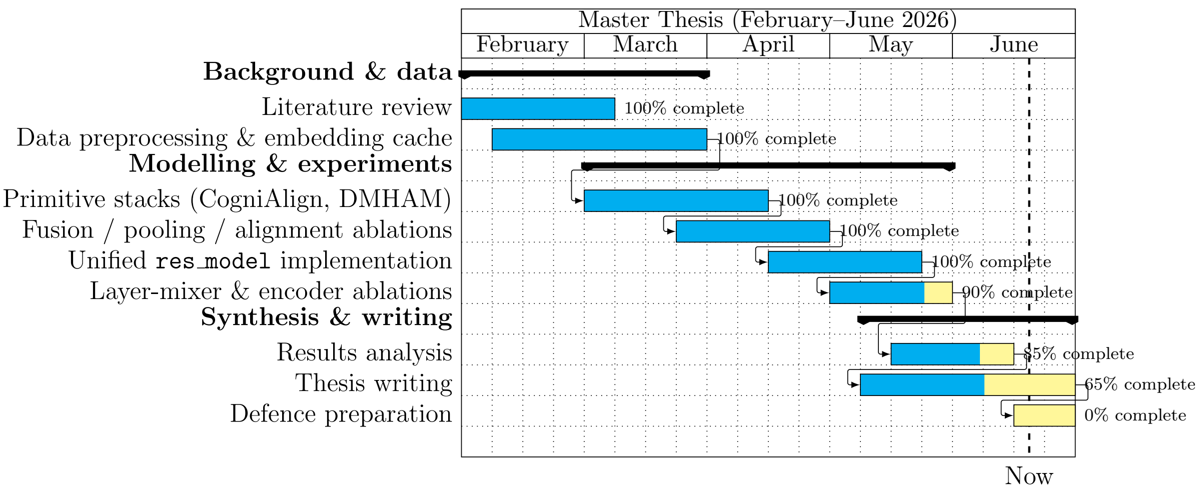

Figure 1: Gantt diagram of the project.

2 State of the art of the technology used or applied in this thesis

This chapter surveys the scientific and technical background that motivates analysis of spontaneous speech with deep multimodal models in neurodegenerative disease. The discussion follows the structure of recent surveys and challenge-led benchmarking in speech-based cognitive assessment [12], but it is organised around the specific threads that underpin the companion contribution developed in this thesis [14]: classical speech markers and the shallow classifiers built on them, handcrafted paralinguistic descriptors, self-supervised acoustic encoders paired with language-model text encoders, fusion architectures that align modalities in time, the two multimodal frameworks (CogniAlign and DMHAM) that instantiate these choices, and the public corpora on which these systems are compared. Throughout, emphasis is placed on results and modelling trends that are reproducible on public benchmarks, notably the Alzheimer's Dementia Recognition through Spontaneous Speech Only (ADReSSo) corpus associated with the challenge formulation of Luz et al. [11], and on how those trends inform deployment on clinically curated cohorts such as the Sant Pau Initiative on Neurodegeneration (SPIN) [13]. The clinical and biological motivation for using speech at all, amyloid pathology and the overlapping primary-progressive-aphasia (PPA) syndromes, is set out in the introduction; here the focus is on the computational methods.

2.1 Speech-based assessment before deep learning

Prospective clinicopathological studies and controlled biomarker programmes increasingly incorporate speech and language markers alongside imaging and fluid assays because voice can be acquired frequently, remotely, and at comparatively low marginal cost [6]. At the same time, evaluation methodology matters: speech measures can track syndrome severity and risk enrichment, yet population heterogeneity (education, dialect, comorbid hearing loss, task instructions) can inflate apparent effects unless modelling assumptions are carefully matched to sampling designs [5].

Early computational work emphasised engineered linguistic descriptors extracted from transcripts (lexical diversity, syntactic complexity, idea density) and shallow acoustic statistics computed from short recordings of spontaneous speech to separate AD from healthy ageing [8]. These approaches demonstrated feasibility but often relied on manual transcription pipelines or handcrafted feature dictionaries whose coverage is incomplete for noisy clinical audio. More recent work revisits classical baselines (random forests, support-vector machines) with richer descriptors, but reports sensitivity that swings with task elicitation and recording conditions [5]. The common thread is that these marker families do carry predictive signal, yet shallow decision boundaries leave the temporal dynamics of the waveform largely untapped [16, 17, 18].

The same shallow-classifier era produced the first quantitative markers for differentiating PPA variants from connected speech, seeking measures that track the breakdown profiles hypothesised by neurolinguistic models: reduced speech rate and grammatical complexity in nfvPPA; lexical-retrieval failures and semantic thinning in svPPA; hesitation and phonological-loop limitations with spared grammar in lvPPA [2]. Empirical studies highlight pause structure, filler usage, lexical diversity, and discourse-coherence metrics as discriminative at group level [19], while articulatory-tempo measures help separate non-fluent from fluent variants [20, 21], and language-aware descriptors discriminate PPA variants well even outside English [22]. At the intersection with amyloid biology, connected-speech features associate with cortical amyloid burden within PPA cohorts [18], reinforcing the notion that speech captures not only syndrome labels but correlates of underlying pathology.

2.2 Handcrafted acoustic descriptors

Despite the strength of the self-supervised waveform encoders introduced below, clinically interpretable descriptors remain attractive because they tie predictions to measurable articulatory and phonatory parameters. These descriptors span the families long used in speech-pathology research, from temporal and pause statistics to prosodic, voice-quality, spectral, and cepstral measures; the concrete 55-feature set this thesis uses, broken down by group, is specified in the methodology (Table 4). Pause-oriented statistics repeatedly emerge as salient in cognitive decline [23, 24]; voice-quality perturbations may track neurogenic speech-motor-control deficits [6].

Open-source dementia-speech pipelines demonstrate competitive accuracy when coupling classical descriptors with modern backbones [25], supporting hybrid designs where auxiliary vectors augment attention pooling or concatenate immediately before classification [14]. Whether explicit features complement latent embeddings in practice depends on architecture and task: pathology-linked cues sometimes persist outside generic waveform embeddings, whereas architectures with explicit alignment may subsume part of the prosodic signal [15].

2.3 Deep learning for speech-based neurodegeneration detection

Representation learning replaces brittle front-ends with hierarchical embeddings trained either supervised end-to-end or semi-supervised with objectives aligned to phonetic/lexical structure.

2.3.1 Self-supervised acoustic encoders: Wav2Vec 2.0

In speech processing broadly, self-supervised pre-training of transformer acoustic models learns frame-level features transferable across languages and domains with comparatively modest labelled downstream data. The Wav2Vec 2.0 architecture [9] illustrates this paradigm. A multi-layer convolutional feature encoder first transforms raw waveform samples into a sequence of latent speech representations at approximately 20-millisecond resolution. These latent vectors are then contextualised by a transformer encoder that attends across the full temporal extent of the utterance. During pre-training, a proportion of the latent frames is masked, and the model solves a contrastive task: for each masked position, it must identify the true quantised representation among distractors sampled from the same utterance. Because no transcriptions or labels are required, the procedure scales to vast quantities of unlabelled audio. The resulting transformer layers learn representations at different levels of abstraction: earlier layers tend to encode acoustic and phonetic detail, while later layers capture more abstract prosodic and linguistic patterns. This layered organisation has practical implications for downstream tasks, which may benefit from aggregating information across multiple transformer layers rather than relying on the final output alone, a question addressed experimentally in this thesis. Two configurations of this model are relevant to the present work. The English `base-960h` model, trained on read English speech, is one; the multilingual XLSR-300M variant, which extends the same framework to around 436,000 hours of speech spanning 128 languages, is the other, and its cross-lingual pre-training makes it better suited to scenarios where labelled pathological speech is scarce in any individual language, such as the Spanish cohorts studied here. The thesis employs both as frozen feature extractors and compares them directly in a controlled encoder ablation, since the more powerful multilingual model is not guaranteed to win once the downstream task is fixed. Related self-supervised models, notably HuBERT [26] and WavLM [27], pursue analogous representation objectives through masked-prediction and denoising tasks respectively, but are not employed in this thesis.

2.3.2 Contextual text encoders

On the text side, the BERT family of encoder-only transformer models [10] learns bidirectional contextual embeddings through masked language modelling: random tokens in the input are masked, and the model is trained to predict them from surrounding context. The resulting representations capture lexical semantics and local syntactic structure, going beyond the static word vectors of earlier approaches. Lighter distilled variants such as DistilBERT [28] preserve much of this capacity in a smaller architecture, while modern encoder-only models such as ModernBERT [29] improve representation quality through updated architectural choices. When automatic speech recognition transcripts are available, these text encoders provide a complementary linguistic view that acoustic features alone may encode only implicitly.

Reviews oriented specifically at AD detection from speech highlight two robust tendencies [12]: deep neural pipelines dominate leaderboard-style benchmarks when adequate training partitions exist, and multimodal systems that consume both acoustic embeddings and text outperform either modality alone when transcripts are informative enough for alignment. These observations motivate architectures that treat spontaneous speech as *tightly coupled* audio–text sequences rather than as isolated vectors [30].

2.3.3 Multimodal fusion

Multimodal classifiers must reconcile wildly different sampling rates: acoustic frames at millisecond resolution versus word or sub-word tokens at comparatively coarse timestamps derived from automatic speech recognition [12]. Three broad paradigms describe how audio and text modalities are combined (Figure 2) [12]. In early fusion, modality embeddings are pooled along the temporal axis and concatenated into a single fixed-length vector before classification. In late fusion, modality-specific classifiers are trained independently and their outputs are combined at the decision level. Intermediate fusion, the paradigm most relevant to this thesis, processes both modalities jointly through sequence-to-sequence layers that operate on the combined representation before the classification head [14].

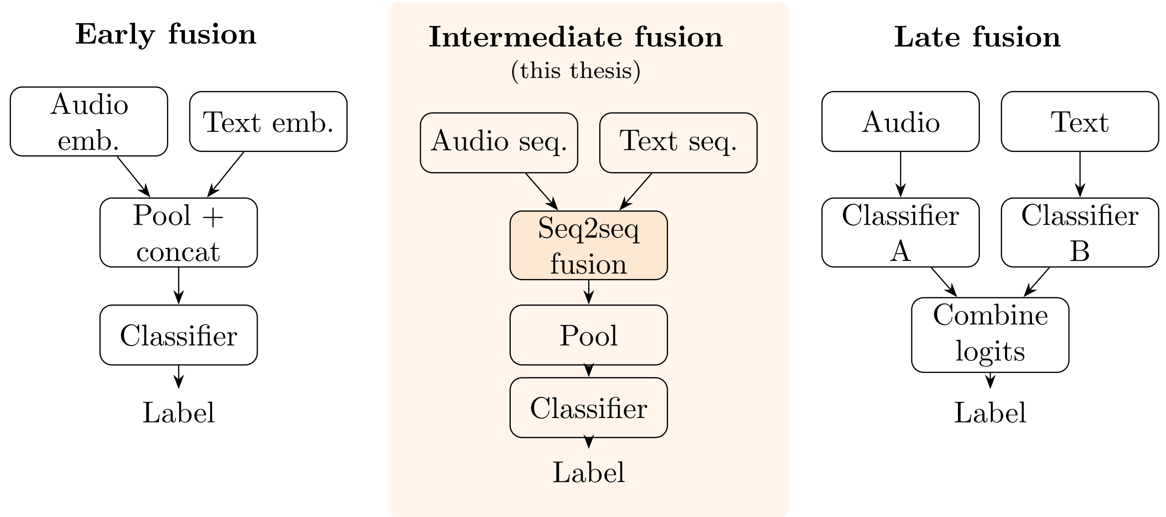

Figure 2: The three multimodal fusion paradigms.

Within intermediate fusion, the seq2seq layer can take several forms, which the thesis later treats as a design axis in its own right; the families below are sketched only at the level needed to position the contribution, and the specific variants and their rationale are developed in the methodology (Section 3). **Self-attention** concatenates both modalities into one sequence and lets every position attend to every other regardless of origin [31], with multi-head and positional-encoding variants adding parallel subspaces and explicit ordering. **Cross-attention** instead introduces directionality, one modality querying the other; a gated variant weights the cross-modal contribution so an unreliable stream can be down-weighted, and a bidirectional variant exchanges information in both directions at once. **Selective state-space models** such as Mamba [32] drop attention altogether, mixing the sequence through a recurrent, input-dependent hidden state in linear rather than quadratic time, which is attractive for the long recordings of spontaneous speech.

2.3.4 Cross-modal alignment

Orthogonal to the choice of fusion mechanism, *cross-modal alignment* addresses the problem of establishing temporal correspondence between streams that operate at fundamentally different resolutions. Acoustic encoders such as Wav2Vec 2.0 produce one contextual frame approximately every 20 milliseconds, whereas text encoders emit one embedding per subword token; a single spoken word may span dozens of acoustic frames while simultaneously being split into multiple subword units by the tokeniser. Without explicit alignment, the fusion layer must learn this correspondence from limited labelled data, risking wasted capacity and spurious associations between acoustically and textually distant positions [15]. The CogniAlign architecture [15] addresses this with a word-to-token alignment strategy (Figure 3). Word-level timestamps are first obtained from the automatic speech recognition step (Whisper), determining which acoustic frames correspond to each spoken word. These word boundaries are then mapped to the subword tokenisation of the text encoder, pairing each textual token with the acoustic frames of the word from which it derives. The aligned representations are then fused by cross-attention. CogniAlign introduces bidirectional cross-attention for this stage, although in its own experiments a gated cross-attention variant reported the strongest result; in either form, temporal synchronisation encourages the network to attend to co-occurring acoustic evidence (such as jitter, spectral tilt, or prolonged closures) alongside the corresponding lexical content rather than to correlates from unrelated temporal positions.

This thesis builds on the CogniAlign alignment idea but generalises it along two independent directions. First, where CogniAlign aligns at the word level and attaches every subword token of a word to the same pooled acoustic span, the present work refines the correspondence to the subword granularity itself, introducing two token-to-token schemes: a *uniform* scheme that divides a word's acoustic frames evenly across its subword tokens, and a *proportional* scheme that allocates them in proportion to each token's character length. Second, where CogniAlign keeps alignment and fusion coupled (pairing its word-level alignment with cross-attention fusion, for which it reports a gated variant as strongest), this thesis decouples the two and treats fusion as a separate axis spanning the families surveyed above. Bidirectional cross-attention, itself one of the fusion mechanisms used by CogniAlign, is adopted as the default of the resulting grid. Both extensions, and their interaction with Wav2Vec layer pooling, are developed in the methodology chapter (Section 3) and evaluated in the results (Section 4).

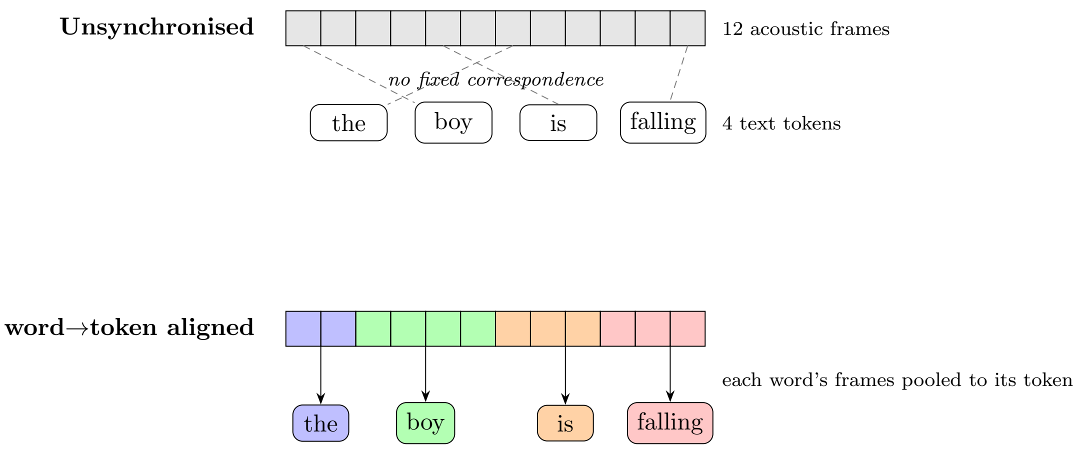

Figure 3: Cross-modal alignment: unsynchronised (top) versus word-to-token aligned (bottom).

2.3.5 Harder tasks and architectural reuse

Beyond binary AD detection, deeper models have been pushed onto the harder multi-class PPA problem: large language models paired with curated linguistic measurements reach high accuracy in retrospective cohorts [33], and architectures trained jointly on acoustic and linguistic embeddings rival expert benchmarks on the variant-classification task [34]. These results stay sensitive to sample size and label noise, and validation across accents, tasks, and cohorts remains an open problem [35]. Fusion architectures developed for speaker and emotion modelling, moreover, have been shown to generalise across paralinguistic tasks once domain regularisation is added [36]; that lineage is the reason this thesis reuses such a multimodal architecture, instantiated with general-purpose frozen speech and text encoders rather than any emotion-trained weights, for clinical speech [14].

2.4 Two multimodal frameworks: CogniAlign and DMHAM

The fusion and alignment ideas above are instantiated in two existing multimodal frameworks that this thesis takes as its starting points: **CogniAlign** [15], a word-to-token aligned pipeline that combines a Wav2Vec 2.0 acoustic stream and a DistilBERT text stream by cross-attention, and **DMHAM**, the Double Multi-Head Attention Multimodal system [36, 14] from the speaker- and emotion-recognition lineage, which forgoes subword alignment and fuses segment-level representations from a cross-lingual XLSR-300M encoder and a ModernBERT text encoder. Rather than treat them as competing monolithic systems, this thesis decomposes them into the independent techniques surveyed above; both frameworks are presented in architectural detail, with their diagram, in the methodology chapter (Section 3.4, Figure 9).

2.5 Benchmarks and datasets

Community challenges accelerated reproducible comparison by freezing splits and evaluation protocols for cognitive-impairment detection from speech-only submissions [11]. The ADReSSo subset, derived from the broader DementiaBank Pitt corpus of picture descriptions, remains the de facto English benchmark for Alzheimer's-dementia versus healthy-control discrimination under matched recording conditions; numerous challenge systems established competitive baselines [37], and subsequent work continues to extend feature attribution and model explanations [25]. DementiaBank itself remains the largest shared resource of this kind, but most of its speech is English and centred on dementia-versus-control labels, which leaves both other languages and pathology-grounded labels (such as amyloid status or PPA variant) comparatively under-served.

Performance on English benchmarks does not carry over to other languages for free. Morphology, prosodic norms, and clinician elicitation protocols all differ, and large labelled neurodegeneration cohorts are scarce outside a handful of reference centres. Initiatives such as SPIN address biomarker discovery and validation by assembling multicentre observational data spanning genetics, imaging, fluid markers, and structured cognitive assessments [13]; subsets incorporating spontaneous speech tasks tied to standard batteries (for example picture-description prompts compatible with the Western Aphasia Battery [7]) enable language-specific baselines that complement English challenge corpora. Spanish-focused studies show that language-aware descriptors can still discriminate PPA variants well [22]. Bilingual analyses add a useful caveat: acoustic models often hold up when a patient speaks in a non-dominant language, whereas lexical models grow brittle [38], a contrast worth keeping in mind for multimodal pipelines deployed in multilingual clinics.

2.6 Synthesis and positioning

Taken together, the literature points to four design commitments reflected in the methodological arc of this thesis [14]: *(i)* evaluate pathology-aware models where biology-grounded labels exist (for example Aβ status alongside syndrome diagnoses), not solely dementia-vs-control proxies; *(ii)* benchmark competing multimodal fusion philosophies using harmonised preprocessing on both an English reference corpus (ADReSSo) and a clinically curated Spanish-speaking subset aligned with SPIN; *(iii)* extend the word-to-token alignment proposed by CogniAlign [15] with two novel token-to-token schemes (uniform and proportional) and evaluate their interaction with Wav2Vec layer pooling in joint ablations; and *(iv)* quantify when handcrafted acoustic descriptors remain complementary after state-of-the-art self-supervised acoustic encoding and explicit audio–text alignment [12, 25].

The following methodology chapter translates these commitments into concrete architectures, training protocols, and evaluation choices.

3 Methodology and Project Development

3.1 Problem Formulation

The general problem addressed in this thesis is the prediction of a clinical label from a single recording of spontaneous speech elicited by a structured picture-description task. Each sample is a pair $(x^{\mathrm{a}}, x^{\mathrm{t}})$, where x^{a} is the raw acoustic waveform and x^{t} is the corresponding transcript of the utterance, and the goal is to learn a mapping $f_\theta : (x^{\mathrm{a}}, x^{\mathrm{t}}) \mapsto y$, where $y \in \mathcal{Y}$ is a discrete diagnostic label whose domain depends on the task.

We study three tasks, each defined over a distinct clinical cohort and summarised in Table 1. The first is Alzheimer's disease classification, a binary discrimination between AD patients and cognitively normal (CN) controls on the English ADReSSo cohort [11], with $\mathcal{Y} = \{\mathrm{AD}, \mathrm{CN}\}$; this is the only task with an official held-out test partition. The second is amyloid-β status prediction, a binary proxy-biomarker task defined on a Spanish-speaking subset of the SPIN cohort [13] elicited with the Western Aphasia Battery picture-description task [7], with $\mathcal{Y} = \{\mathrm{positive}, \mathrm{negative}\}$. The third is primary progressive aphasia variant classification, a three-class problem distinguishing the non-fluent/agrammatic, semantic, and logopenic variants [2] defined on the same SPIN subset, with $|\mathcal{Y}| = 3$. Both Spanish tasks are drawn from this single SPIN subset and share the same data tree; neither has an official external test split, so for these we report cross-validated performance only and never construct an artificial test partition.

Both modalities are encoded by frozen self-supervised models. The acoustic stream is processed by a Wav2Vec 2.0 encoder [9] (the specific variant depends on the implementation and is detailed in Section 3.3.1) producing a frame-level sequence $\mathbf{H}^{\mathrm{a}} = (\mathbf{h}^{\mathrm{a}}_1, \ldots, \mathbf{h}^{\mathrm{a}}_{T_a})$, and the textual stream by a frozen transformer text encoder of the BERT family [10], producing a token-level sequence $\mathbf{H}^{\mathrm{t}} = (\mathbf{h}^{\mathrm{t}}_1, \ldots, \mathbf{h}^{\mathrm{t}}_{T_t})$. Because the two sequences live on different time bases ($T_a \neq T_t$), an alignment step is required to place acoustic and textual representations on a common index before they can be fused, after which a sequence-to-sequence fusion module combines the aligned streams into a representation that feeds a shallow classification head.

Rather than treating this pipeline as a single monolithic system optimised end to end, this thesis formulates it as the composition of six largely independent design axes, each of which can be varied while the others are held at controlled defaults. The modality axis m determines whether the classifier consumes audio alone, text alone, their combination, or additionally a set of handcrafted acoustic features. The fusion axis f selects the sequence-to-sequence architecture that combines the aligned acoustic and textual streams, drawn from a catalogue of seven families. The alignment axis a fixes the scheme that maps acoustic frames onto text positions, comprising the word-to-token baseline of CogniAlign [15] and the two token-to-token schemes proposed in this work. The Wav2Vec pooling axis w chooses between the final encoder layer and a learnable weighted sum across layers. The layer-mixing axis ℓ refines that choice: in place of a single global set of layer weights, an adaptive mixer can learn to combine the encoder's layers per utterance, or even per token, so that the depth of representation is selected dynamically rather than fixed in advance. Finally, the feature-extractor axis e varies the pretrained encoders themselves, substituting one audio or text backbone for another, to test how far the downstream trends depend on the specific self-supervised models that supply the two streams.

A concrete system instance is thus parameterised by a tuple (m, f, a, w, ℓ, e), and the central empirical question is to estimate the marginal contribution of each axis, alone and in interaction, on the three tasks, rather than to identify a single best-performing run. The unit of comparison is therefore the *technique*, not the implementation: two independent codebases replicate the fusion and Wav2Vec-pooling trends under a second software stack, and a third, configuration-unified implementation then consolidates the resulting techniques into a single model (Section 3.4.1). Building on this decomposition, the experimental programme of Section 4 traverses the axes in a staged sequence, and the synthesis of the strongest choice along each axis into a recommended hybrid architecture is taken up in the conclusions (Section 5).

3.2 Datasets and Preprocessing

The three tasks defined in Section 3.1 are evaluated on two clinical cohorts, summarised in Table 1: the English ADReSSo benchmark for AD classification, and a curated subset of the Spanish-speaking SPIN cohort that supplies both the amyloid-β and PPA tasks. They differ in language, label space, and the availability of a held-out test partition, which in turn governs the evaluation protocol applied to each (Section 3.8). All recordings originate from structured picture-description tasks, so the input distribution is comparable across cohorts even though the underlying populations and elicitation languages differ. The cohort descriptions and demographic characterisation below follow the companion paper in which we first introduced this SPIN subset [14]; the remainder of the subsection details each cohort and the common preprocessing pipeline that turns raw recordings into the paired acoustic and textual representations consumed by the model.

Table 1: Clinical cohorts, languages, label spaces, and evaluation protocol. Only the English ADReSSo cohort provides an official held-out test partition; the two Spanish tasks are evaluated by stratified cross-validation. Both Spanish tasks are drawn from the same SPIN subset, elicited with the Western Aphasia Battery picture-description ("Picnic Scene") task.

Task	**Cohort**	**Language**	**Classes**	**Evaluation**
AD vs. CN	ADReSSo	English	2	Official `test-dist` (`test_acc`)
Amyloid-β status	SPIN	Spanish	2	5-fold CV (validation only)
PPA variant	SPIN	Spanish	3	5-fold CV (validation only)

3.2.1 ADReSSo Corpus

The Alzheimer's disease task uses the ADReSSo-IS2021 corpus introduced for the corresponding Interspeech challenge [11]. The corpus is derived from the English-language DementiaBank Pitt Corpus and comprises 237 spontaneous picture-description recordings, totalling approximately 166 minutes of speech, in which participants describe the Cookie Theft scene from the Boston Diagnostic Aphasia Examination. It is balanced between AD patients and healthy controls and matched for age and sex, which removes the most obvious demographic confounds and makes it a clean benchmark for isolating speech-based effects. The challenge distributes a fixed train and `test-dist` split, so this is the only task in the thesis with an official external test set and therefore the only one on which we report a held-out test accuracy in addition to cross-validated performance. We use ADReSSo primarily to anchor the architecture against a well-documented English benchmark, as in our companion paper [14], rather than to optimise exhaustively for the challenge leaderboard.

3.2.2 SPIN Clinical Cohort

The two Spanish tasks are defined on a curated subset of the Sant Pau Initiative on Neurodegeneration (SPIN) [13], a longitudinal cohort for biomarker discovery and validation in neurodegenerative disorders maintained at the Hospital de la Santa Creu i Sant Pau in Barcelona. From this cohort we assembled a set of Spanish-speaking participants for whom spontaneous speech recordings, a confirmed amyloid-β status, and, where applicable, a clinical PPA variant diagnosis were all available [14]. Unlike ADReSSo, the elicitation stimulus is the "Picnic Scene" from the Western Aphasia Battery [7], and recordings were captured during routine clinical evaluations, with a variable number of samples per subject ranging from one to seven. This repeated-measures structure is the reason all cross-validation on SPIN is stratified by subject identifier, so that recordings from the same participant never straddle the train and validation folds; reporting at the recording level rather than the subject level would otherwise leak speaker identity and inflate the estimated accuracy.

The subset supports two clinically distinct tasks. The amyloid-β prediction task is a binary problem that asks the model to predict amyloid positivity, a fluid biomarker of AD whose confirmation otherwise requires costly or invasive cerebrospinal-fluid assays or PET imaging. At the subject level it comprises 53 amyloid-positive and 49 amyloid-negative participants, and at the recording level 114 positive and 98 negative samples, leaving the task approximately balanced. The PPA variant classification task is a three-class problem distinguishing the logopenic (lvPPA), non-fluent/agrammatic (nfvPPA), and semantic (svPPA) variants, with 35, 22, and 17 subjects and 104, 63, and 35 recordings respectively. This task is markedly imbalanced (svPPA accounts for fewer than a third of the lvPPA recordings), and the small per-class subject counts make fold-level performance estimates high-variance, so the PPA results are interpreted with corresponding caution.

Tables 2 and 3 report the demographic composition of the two tasks. No statistically significant differences in age, sex, or years of education were found across the groups of either task, assessed with the Wilcoxon rank-sum test for continuous variables and Pearson's chi-squared test for categorical ones [14]; this matching is important because it makes it less likely that a classifier exploits demographic shortcuts rather than disease-related speech characteristics.

Table 2: Demographic composition of the SPIN amyloid-β task by subject. Values are mean $\pm$ SD or n (%).

	Negative ($N = 49$)	**Positive** ($N = 53$)
Age (years)	72.8 ± 6.1	72.6 ± 6.8
Female	25 (51.0%)	22 (41.5%)
Male	24 (49.0%)	31 (58.5%)
Education (years)	12.8 ± 6.6	11.8 ± 5.7

Table 3: Demographic composition of the SPIN PPA task by subject. Values are mean $\pm$ SD or n (%).

	lvPPA ($N = 35$)	**nfvPPA** ($N = 22$)	**svPPA** ($N = 17$)
Age (years)	72.6 ± 6.5	74.7 ± 4.9	73.4 ± 5.8
Female	15 (42.9%)	9 (40.9%)	11 (64.7%)
Male	20 (57.1%)	13 (59.1%)	6 (35.3%)
Education (years)	10.4 ± 6.4	12.6 ± 5.9	9.8 ± 6.9

3.2.3 ASR Pipeline and Transcript Quality

The textual modality is derived from the audio rather than from gold manual transcripts, so that the pipeline reflects a realistic deployment in which only recordings are available. Each recording is first truncated to its initial forty seconds of speech, which approximates the effective input length of the alignment-based model and, on the SPIN data, avoids capturing the clinician speech that occasionally appears toward the end of an evaluation. The truncated audio is then transcribed by automatic speech recognition (ASR), using Whisper, though the two stacks differ in configuration: the CogniAlign pipeline runs the Whisper `turbo` model offline to obtain transcripts together with the word-level timestamps its alignment step requires, whereas the DMHAM pipeline reuses precomputed transcripts where they are available and falls back to the lightweight Whisper `tiny` model otherwise. The resulting transcript is tokenised by the text encoder before being aligned with the acoustic stream. Because recognition errors propagate into the textual representation, transcript quality is a genuine source of variance, particularly for the Spanish cohort and for the disfluent or aphasic speech characteristic of PPA; the experiments therefore treat the text modality as one ablatable input among several rather than as a noise-free signal.

3.3 System Modules

This section describes the building blocks of the pipeline and the options available for each, taken in the order data flows through the model: the audio and text branches that embed each stream into a sequence of vectors, the handcrafted acoustic features that enter as a third input, the alignment-and-fusion stage that places the two streams on a common index and combines them, and the shallow head that reads out a label. How a concrete system wires these modules together, and the three implementations in which they are realised, is taken up in the system-architecture section that follows (Section 3.4).

3.3.1 Audio Branch: Encoder and Layer Pooling

Audio is encoded by a frozen Wav2Vec 2.0 model [9], but the two stacks use different variants by design. CogniAlign uses the base configuration (`facebook/wav2vec2-base-960h`): a convolutional feature encoder maps the raw 16 kHz waveform to latent frames at roughly 20 ms resolution (about 50 frames per second), which a twelve-layer transformer context network then contextualises into 768-dimensional hidden states. This model has on the order of 95 million parameters and was trained on 960 hours of English read speech (LibriSpeech); it is therefore monolingual, and CogniAlign applies it unchanged to the Spanish cohorts as well. DMHAM instead uses the cross-lingual XLSR-300M variant, a deeper encoder of roughly 300 million parameters whose twenty-four transformer layers produce 1024-dimensional states and which was pretrained on about 436,000 hours of unlabelled speech spanning 128 languages. The two audio encoders thus differ in depth, width, capacity, and language coverage.

Within this same audio branch, how the frame-level sequence is read out of the encoder's stack of layers is itself a design axis, the Wav2Vec-pooling axis. The encoder is used purely as a frozen feature extractor: its weights are never updated, and the sequence $\mathbf{H}^{\mathrm{a}}$ is taken directly from the transformer stack. The default, used as the controlled setting in every phase except the dedicated pooling study, takes the hidden states of the final transformer layer. The alternative learns one scalar weight per layer, normalises the weights with a softmax, and forms the representation as the weighted sum of all layer outputs (twelve layers for the base encoder in CogniAlign, twenty-four for XLSR-300M in DMHAM). This is motivated by the well-documented observation that the layers of a self-supervised speech encoder specialise in different information (lower layers carry phonetic and speaker detail while higher layers carry more abstract content), so a learnable combination can expose cues that any single fixed layer discards. Acoustic data augmentation is not used in the reported runs, which rely on the clean precomputed embeddings introduced in the training protocol; whether augmentation helps within this cached, frozen-encoder setting is examined separately in Section 3.6.

The weighted sum just described shares a single set of layer weights across every recording. Making that combination learnable has a clear purpose: instead of committing in advance to the final layer, the model decides for itself which layers to read features from, putting weight wherever the task-relevant signal is strongest. In the recommended base configuration the weighting is a learnable softmax over all twelve transformer layers, and over all twenty-four for the deeper XLSR-300M encoder.

The most informative depth, however, need not be the same for every recording. A refinement evaluated in the unified `res_model` implementation therefore makes the weighting adaptive, replacing the one fixed vector with a learnable *layer mixer* that forms the softmax over the layer logits conditionally. It comes in three granularities. A *global* mixer keeps one static weighting for all data. A *sample* mixer predicts a separate weighting per utterance, letting an atypical or noisy recording lean on different layers. A *token* mixer goes finer still, mixing the layers independently at each position. A low temperature ($\tau = 0.1$) sharpens the softmax so the mixer settles on a few layers rather than smearing weight across all of them, and its parameters are trained at about ten times the rate of the rest of the head. The sample mixer is the one carried into the recommended configuration: lightweight, input-conditioned, and in practice biased toward the upper layers.

Mechanically, the mixer acts only on the audio branch and only before fusion, collapsing the layer axis into a single vector per time step; the text embeddings never pass through it. The sample mixer obtains its per-utterance weights by mean-pooling each layer over time and passing the result through a small linear scorer followed by the softmax, so the weighting reflects global properties of the recording; the token mixer instead scores every layer at each position, which amounts to a lightweight attention over layers. This layer mixing should not be confused with attention pooling: the mixer reduces the *depth* (layer) axis before fusion, whereas attention pooling would reduce the *time* axis after fusion, and the recommended configuration uses simple mean pooling for the latter.

3.3.2 Text Branch: Encoder

Text is encoded by a frozen transformer from the BERT family [10], which turns the tokenised transcript into a sequence of contextual token-level embeddings $\mathbf{H}^{t}$. The two implementations again use different members of this family. CogniAlign uses DistilBERT [28], a six-layer encoder distilled from BERT-base that retains most of its language-understanding capability at roughly half the parameters and inference cost. DMHAM uses ModernBERT-base [29], a more recent encoder of about 149 million parameters trained on the order of two trillion tokens, with a native 8192-token context window, rotary positional embeddings, GeGLU activations, and an alternating local–global attention pattern that makes long-context encoding efficient.

As with the acoustic encoder, the text encoder is frozen and its embeddings are precomputed. Holding it fixed within each stack ensures that the text representation is not a confound when fusion, alignment, or pooling is varied.

3.3.3 Handcrafted Acoustic Features

In addition to the learned embeddings, the pipeline can incorporate a set of 55 utterance-level handcrafted acoustic descriptors designed to capture fine-grained paralinguistic cues that are clinically interpretable but not easily recovered by a self-supervised encoder operating on the raw waveform. They are produced by a feature-extraction script adapted from NeuroXVocal [25], which demonstrated the set's effectiveness on the ADReSSo benchmark, and were extended to the SPIN cohort in our companion paper [14]. Extraction relies on `librosa` together with Praat-based analysis via `parselmouth`; the resulting vector is computed once per recording and stored, so that the identical 55-feature set is reused across every experiment that enables it. The descriptors span the five groups listed in Table 4: temporal and pause statistics, acoustic–prosodic measures, voice-quality measures, spectral descriptors, and cepstral coefficients (mel-frequency cepstral coefficients, MFCCs).

Table 4: The five groups of handcrafted acoustic descriptors (55 features in total), extracted with `librosa` and Praat (`parselmouth`).

Group	Descriptors
Temporal & pause	Total speech duration, speech-to-pause ratio, number of pauses, pause-duration statistics (mean, maximum, SD)
Acoustic–prosodic	Pitch (mean, SD, range), intensity, speaking rate, articulation rate
Voice quality	Local jitter, local shimmer, harmonics-to-noise ratio
Spectral	Spectral-centroid statistics, zero-crossing rate, formants F1–F3 (mean, SD)
Cepstral	First 13 MFCCs (mean, SD)

The five groups are grounded in the clinical phenomenology of the target disorders. Pause behaviour is the clearest case: a higher pause rate, longer pauses, and a lower speech-to-pause ratio are well-established markers of Alzheimer's disease, where they track the lexical-retrieval and speech-planning difficulties that accompany cognitive decline. Voice quality plays a complementary part, with jitter and shimmer registering the subtle phonatory instabilities that can appear before any overt symptom. Such descriptors earn a separate place for two reasons: they are explicit and clinically interpretable, and a self-supervised encoder offers no guarantee of preserving them. Pretrained to recover phonetic and lexical content, the encoder has no objective that obliges it to retain absolute timing or low-level voice-quality statistics, so it may discard precisely the cues a clinician would attend to.

Before the features are used they are normalised, and the normalisation is computed with care to avoid leakage: each dimension is Z-score standardised using only the mean and standard deviation of the training split of the fold in question, never statistics pooled over the whole dataset.

Integration follows a late-fusion design. The normalised 55-dimensional vector is first projected to the model's 768-dimensional hidden width, then combined with the multimodal stream at one of two late points, prepended as an additional token to the sequence that enters the fusion module, or concatenated to the already-pooled fixed-size embedding immediately before the classification head. Either way the descriptors join only after the self-supervised streams have been formed, so the fusion and cross-attention machinery still operate on the learned representations alone and the handcrafted cues act as an interpretable side-channel rather than something the attention has to disentangle from the waveform.

The descriptors are ablated explicitly in the modality study, where the audio-plus-text-plus-features cell is compared against the audio-plus-text cell so that any contribution of the handcrafted features is measured against an otherwise identical model. Beyond that controlled comparison, the CogniAlign ablation grids keep the 55-feature vector enabled by default whenever the cohort's feature CSV is available, so the fusion, alignment, and pooling studies are run on top of the audio-plus-text-plus-features input rather than with the features removed.

3.3.4 Cross-Modal Alignment and Fusion

Once both streams are embedded (and the handcrafted features prepared as a third input), they must be placed on a common sequence index before they can be combined; this module covers both stages, alignment then fusion. Alignment is required because the acoustic sequence is far longer than the token sequence ($T_a \gg T_t$). CogniAlign resolves this by precomputing an aligned tensor in which each text position is paired with a pooled acoustic representation, and the three alignment schemes differ in how acoustic frames are assigned to text positions. The word-to-token baseline, inherited from CogniAlign [15], uses the word-level timestamps produced by the ASR step to mean-pool the acoustic frames spanned by each spoken word into a single vector, which is then replicated at every subword token of that word. The two schemes proposed in this thesis operate one level finer, at the subword granularity produced by the text tokeniser. The *uniform* scheme takes the frame span of each word and divides it evenly among the word's subword tokens, so that a word split into three tokens contributes three equal slices. The *proportional* scheme instead divides the span in proportion to the character length of each subword token, on the assumption that longer subwords tend to occupy more of the spoken word and should therefore receive proportionally more acoustic frames. Both subword schemes yield a finer-grained audio–text correspondence than the word-level baseline, and the comparison among the three constitutes the alignment axis (Figure 4). DMHAM does not align at the subword level: it fuses segment-level representations and therefore takes no part in the alignment ablation.

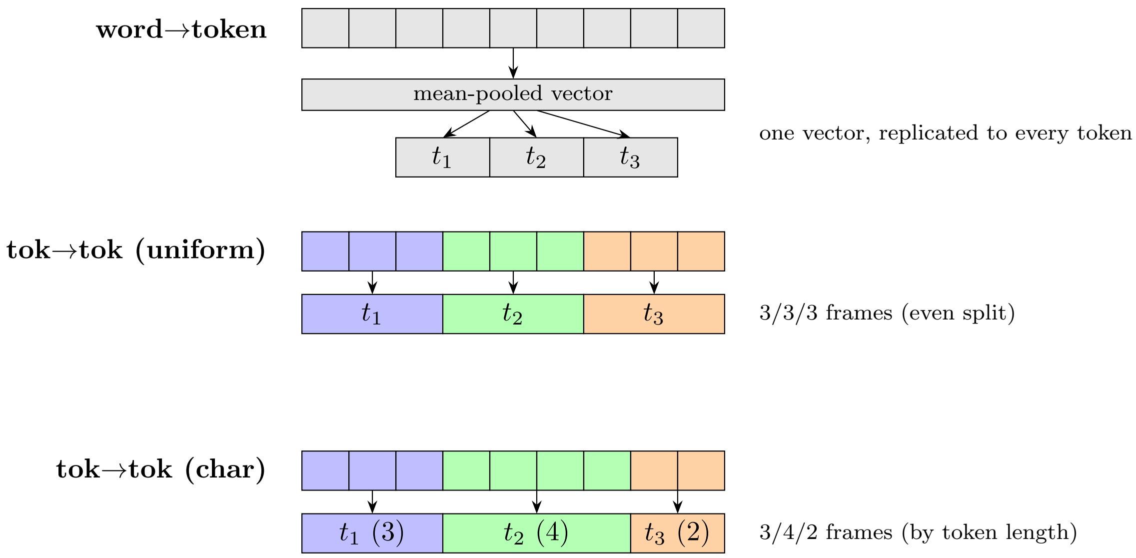

Figure 4: The three audio–text alignment schemes compared in this thesis.

With the streams aligned (or, in DMHAM, taken at the segment level), the second stage of this module is fusion: it combines them into the joint representation that the head consumes. This is the axis the thesis varies most widely. Seven families are implemented and compared, summarised in Table 5, and they fall into three groups (Figures 5–7) that embody three distinct hypotheses about how acoustic and textual evidence ought to interact.

The first group applies **self-attention** to the concatenated audio–text sequence, treating both modalities as one undifferentiated set of tokens in which every position may attend to every other. Its premise is that the relevant cross-modal structure will surface on its own once the model is given global context, with no architectural separation between the streams. Three variants probe how much machinery that premise actually needs. Plain self-attention uses a single attention map. Multi-head self-attention splits the computation into several parallel heads, each free to specialise in a different relation, one head might track how prosody aligns with content words while another tracks how pauses fall around syntactic boundaries. The transformer-with-positional-encoding variant goes further and injects sinusoidal position information inside a full transformer block [31], so that the order of frames and tokens is modelled explicitly. That last addition matters for speech in particular, where timing is itself diagnostic and a permutation-invariant model would throw away exactly the rhythm that distinguishes fluent from disfluent production.

[Figure to be added]

Figure 5: Self-attention fusion over the concatenated audio–text sequence.

The second group, **cross-attention** (Figure 6), keeps the modalities apart and lets one query the other. One stream forms the queries while the other supplies the keys and values, so the fused representation is essentially one modality re-expressed in terms of the most relevant parts of the other. Gated cross-attention extends this with a learned gate that scales the cross-modal contribution, giving the model an explicit means to down-weight the other stream when it is unreliable. That control is not cosmetic here: the text stream is the output of an error-prone ASR step, and a fixed cross-modal coupling would force the model to attend to transcription noise it would be better off ignoring. Bidirectional cross-attention performs the exchange in both directions at once, refining the audio with the text and the text with the audio instead of designating either as the primary stream.

The final family abandons attention altogether. The **Mamba** block [32] is a selective state-

[Figure to be added]

Figure 6: Cross-attention fusion variants: (a) standard, (b) gated, and (c) bidirectional.

space model that mixes the sequence in linear rather than quadratic time, and it carries a very different inductive bias: information is propagated through a recurrent hidden state whose dynamics are themselves input-dependent, which favours smoothly evolving temporal structure over the all-pairs comparison that attention performs. Including it asks whether any gain attributed to fusion is specific to attention or survives under a fundamentally different sequence model.

The same seven families are implemented in both codebases. Fixing the catalogue this way is what lets the fusion axis be read as a property of the technique itself, reproducible across two independent software stacks, rather than an artefact of one implementation.

Two non-sequential baselines accompany the seven families as a floor: plain concatenation and an element-wise mean of the two pooled stream embeddings, which combine the modalities without any cross-modal interaction. They are not sequence-to-sequence fusions and so are not counted among the seven, but they mark the lower bound the families must clear and are reported alongside them in the fusion results (Table 9).

[Figure to be added]

Figure 7: Mamba selective state-space fusion applied to the concatenated audio–text sequence.

Table 5: The seven sequence-to-sequence fusion families compared in the experiments, grouped by mechanism.

Group	**Fusion family**	**Core mechanism**
Self-attention	Self-attention	Single attention map over the concatenated sequence
	Multi-head self-attention	Parallel attention heads over the concatenated sequence
	Transformer + PE	Transformer block with explicit positional encoding
Cross-attention	Cross-attention	One modality queries the other
	Gated cross-attention	Cross-attention with a learned gate on the cross-modal signal
	Bidirectional cross-attention	Cross-attention exchanged in both directions
State-space	Mamba	Selective state-space mixing in linear time

3.3.5 Classification Heads

The fused sequence is reduced to a single fixed-size vector by a pooling operation (mean pooling in CogniAlign and learned attention pooling in DMHAM), and this vector is passed to a small multi-layer perceptron that outputs the class logits, trained with a cross-entropy objective. The head is kept intentionally shallow so that the bulk of the modelling capacity lies in the frozen encoders and the fusion module under study; this keeps the trainable parameter count low and makes the comparison between fusion families fair, since differences in accuracy can then be attributed to the fusion mechanism rather than to head capacity. The output layer follows the task, with two units for the binary AD and amyloid-β tasks and three for PPA variant classification. All three targets in this thesis are categorical, so the head is always a classifier; the same design would extend directly to a regression target (for instance a continuous amyloid burden rather than a binary status) by replacing the final layer with a single linear unit and the cross-entropy objective with a regression loss, a possibility noted here only to delimit the scope of the present work.

3.4 System Architecture

The model follows a two-stream design in which audio and text are embedded independently by frozen self-supervised encoders, aligned onto a common sequence index, combined by a sequence-to-sequence fusion module, and finally pooled and mapped to a task label by a shallow head (Figure 8; the frozen encoders are drawn in blue, the trainable fusion module in orange, and the optional handcrafted-feature branch in teal, with the features joining either as an extra fusion token or by concatenation after pooling). Each stage is one of the modules described in Section 3.3 and corresponds to one of the design axes introduced in Section 3.1: the encoders and their pooling fix the acoustic and textual representations, the alignment step determines how the two time bases are reconciled, and the fusion module determines how the aligned streams interact. The ablation programme depends on this separation: a single axis can be changed while the rest of the pipeline stays fixed.

Both encoders are kept frozen throughout. This is a modelling choice, not a computational shortcut: freezing holds the trainable parameter count in the low millions, concentrates learning capacity in the fusion module under study, and reduces the risk of overfitting the small clinical cohorts. It has a practical payoff as well. Once the encoder outputs are deterministic, every embedding can be computed once and cached, and that cache is what makes a multi-axis grid of experiments tractable at all.

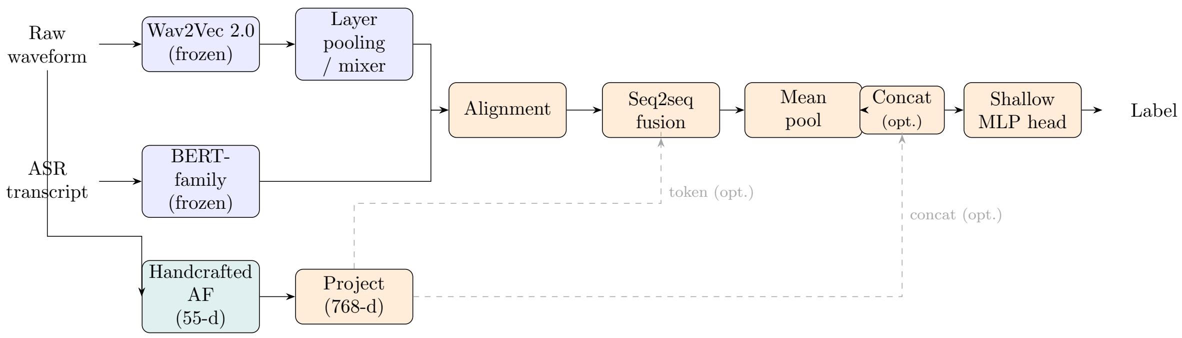

Figure 8: End-to-end two-stream architecture used in this thesis.

The architecture is realised across three complementary implementations, two of which are the existing multimodal frameworks this thesis adopts and decomposes (Figure 9). **CogniAlign** [15] is the word-to-token aligned pipeline: Whisper supplies word-level timestamps, a frozen Wav2Vec 2.0 base encoder and a DistilBERT text encoder produce the two streams, each word's acoustic frames are mean-pooled and attached to its subword tokens, and the aligned streams are combined by cross-attention before a mean-pooled head. **DMHAM**, the Double Multi-Head Attention Multimodal system [36, 14] (distinct from the multi-head self-attention fusion family described above), forgoes subword alignment altogether: it fuses segment-level representations from a cross-lingual XLSR-300M encoder and a ModernBERT text encoder and reads them out with attention pooling. The two share the two-stream skeleton of Figure 8 but differ in their encoders, their ASR, and most consequentially in whether they align at the subword level at all. They serve as the *primitive* stacks in which the individual techniques were developed and ablated: CogniAlign is the full pipeline and the only stack that implements subword token alignment, whereas DMHAM operates on segment-level representations and so serves to check whether the fusion and Wav2Vec-pooling trends reproduce under a second, independent implementation. Because the two differ in encoders, ASR, and training schedule, results are compared across them as directional agreement, not exact reproduction; this caveat applies wherever cross-stack figures are quoted in what follows. The third implementation, `res_model`, consolidates the techniques into a single configuration-driven model from which the recommended architecture is drawn; it is competitive with the primitive stacks across the three tasks without being uniformly the best on any single one, its value lying in integration and deployment rather than in a higher accuracy ceiling. It is described next.

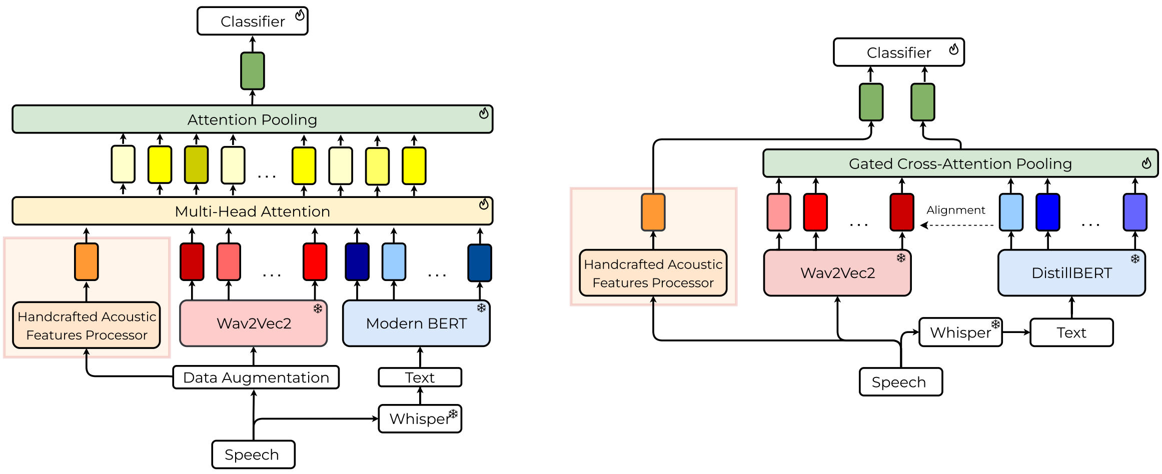

Figure 9: The two frameworks compared: DMHAM (left) and CogniAlign (right).

3.4.1 Unified Implementation: `res_model`

The two primitive stacks established the trends of the individual axes, but each carries incidental design decisions, such as a fixed ASR model, alignment granularity, and training schedule, that make some comparisons impossible to isolate within a single stack. To remove these confounds and to assemble the strongest per-axis choices into one system, the techniques are re-implemented in `res_model`, a configuration-driven trainer in which the modality, fusion family, alignment scheme, Wav2Vec pooling and layer mixer, and the audio and text encoders are all selected from a single configuration. Its results are reported by technique rather than by stack identity, since a given configuration no longer corresponds to “CogniAlign” or “DMHAM” but to an explicit combination of choices. This unified implementation serves three purposes: it reproduces the main technique trends under one codebase, so that they no longer depend on cross-stack differences; it enables the controlled single-factor studies that the primitive stacks cannot express cleanly, namely the adaptive layer mixer of Section 3.3.1 and the encoder ablation of Section 3.4.2; and it provides the vehicle for the recommended architecture.

For the per-task ablations reported in Section 4, `res_model` holds a single *stable recipe* and varies one axis at a time: a learnable weighted sum of Wav2Vec layers, mean time-pooling, bidirectional cross-attention fusion, and no handcrafted features, with the alignment left at each task's default. This is intentionally not the recipe of the legacy CogniAlign grids, whose controlled defaults instead read the final Wav2Vec layer and keep the handcrafted features enabled. The two programmes are read for their per-axis trends rather than for numerically identical cells, and every results table states the recipe behind it.

That recommended architecture is the configuration the ablations point to, anchored on the English ADReSSo task: word-timestamped transcripts and a DistilBERT text encoder with the Wav2Vec 2.0 base audio encoder, the audio read as a weighted layer stack combined by the per-utterance sample mixer, word-to-token alignment, bidirectional cross-attention fusion, and a mean-pooled shallow head, with the handcrafted acoustic features off by default (Figure 10). It is chosen as the strongest configuration that stays stable across folds and seeds, not the single highest-scoring cell of any one run; on the Spanish cohorts the cross-lingual XLSR-300M audio encoder can be preferable, so the stack is read as task-anchored rather than universal. Its accuracy and the synthesis across the design axes are taken up in the conclusions (Section 5).

[Figure to be added]

Figure 10: The unified `res_model` architecture and its recommended configuration.

3.4.2 Feature Extractors: Single-Factor Encoder Ablation

The choice of pretrained backbone for each stream is itself a design axis, the feature-extractor axis introduced in Section 3.1. The two primitive stacks already pair different encoders, but they also differ in ASR, alignment granularity, and training schedule, so reading the gap between them as an encoder comparison would confound the encoder with everything else that varies. A clean comparison requires changing one encoder at a time while holding the rest of the pipeline fixed, which is what the unified `res_model` implementation provides through a single-factor extractor ablation. Against a fixed recipe (bidirectional cross-attention fusion, word-to-token alignment, the weighted Wav2Vec stack with the per-utterance sample layer mixer, and word-timestamped CogniAlign transcripts), one component is swapped per run and every other factor is held constant. On the audio side the base encoder is replaced by XLSR-300M, to test whether a deeper cross-lingual model helps the Spanish cohorts; on the text side DistilBERT is replaced by ModernBERT and by the undistilled BERT-base [10], contrasting a lightweight distilled encoder, a recent long-context one, and the original reference. Each variant is run on all three tasks under two random seeds to expose seed sensitivity. The candidates are deliberately drawn from different points of the design space: DistilBERT for its low cost and its role in the original CogniAlign design, ModernBERT for its modern architecture and long context, BERT-base as the undistilled baseline, and the English base versus cross-lingual XLSR audio encoders for the monolingual-versus-multilingual contrast that is most relevant once the same pipeline is applied to Spanish speech. The per-task outcomes, which prove markedly task-dependent, are reported with the experiments.

3.5 Training Protocol

All models are trained under stratified five-fold cross-validation applied independently to each dataset, so that every fold preserves the class balance of its cohort. The two implementations use schedules tuned to their respective encoders. CogniAlign is optimised with a learning rate of 2×10^{-5} for up to two hundred epochs with early stopping on the validation fold using a patience of fifteen epochs, while DMHAM uses a learning rate of 5×10^{-5} for twenty-five epochs. To separate genuine effects from initialisation noise on these small cohorts, every `res_model` configuration is trained under two fixed random seeds (43 and 3407) and both runs are reported, so that the stability of a result across seeds can be judged alongside its mean. To keep training and inference tractable with frozen encoders, embeddings are precomputed and cached: CogniAlign stores token-aligned tensors as precomputed `.pt` files, and DMHAM maintains an embedding cache. This caching was added to DMHAM after its initial implementation, so all reported inference times correspond to the final cached configuration.

3.6 Cached Audio Augmentation

The reported runs use the clean precomputed embeddings just described. To test whether data augmentation helps *within* this frozen-encoder, cached protocol rather than by abandoning it, an offline cached augmentation scheme was implemented and evaluated on the recommended per-task configurations. Each training waveform is augmented $K = 5$ times offline with additive background noise, reverberation, and time masking; speed perturbation is excluded because it changes the utterance duration and breaks the token alignment. Every augmented view is encoded once by the frozen Wav2Vec 2.0 and aligned with the existing token metadata, then stored as an additional cached tensor, so no re-encoding or re-transcription happens during training. At train time each utterance loads, with probability p_{aug}, one randomly chosen augmented view, while validation and test always use the clean view; only the audio stream is augmented, with the text and handcrafted-feature streams left clean. The protocol is therefore identical to the clean pipeline except for which audio view is sampled during training, and the mixing probability p_{aug} controls how much augmentation the model sees. Its effect is reported in Section 4.4.

3.7 Ablation Protocol

The experiments isolate the contribution of each design axis by varying one axis at a time while the others are pinned to controlled defaults, so any change in performance can be attributed to the manipulated axis. In the CogniAlign grids those defaults are the final-layer Wav2Vec representation, word-to-token alignment, and bidirectional cross-attention fusion, with audio-plus-text input and the 55 handcrafted acoustic features kept on whenever the cohort's feature CSV is available. The modality study is the exception: it is run separately, on a model of its own, to isolate the inputs rather than to compare fusions. Every cell passes through one fixed, symmetric architecture: each enabled modality is linearly projected to a common 768-dimensional width and tagged with a learnable modality-type embedding, the resulting tokens are concatenated behind a shared `[CLS]` token, and a single multi-head self-attention block with a residual connection and layer normalisation mixes them; the `[CLS]` representation is read out by a shallow head. The handcrafted features, when present, enter as one additional projected token carrying the acoustic modality embedding. The audio is read from the final Wav2Vec layer and no alignment is applied, so that only the set of input tokens changes from cell to cell. Because this neutral setup omits the weighted layer pooling and per-task alignment of the later studies, its absolute accuracies are not directly comparable with them and are read within the modality table alone.

The programme runs as a staged sequence. The modality study comes first, then the fusion study; the alignment study compares the three schemes under a fixed fusion (bidirectional cross-attention or Mamba) at the final Wav2Vec layer; the layer-pooling study contrasts the final layer with a learnable weighted sum; and a combined alignment–pooling study crosses the alignment schemes with weighted pooling at bidirectional cross-attention. The strongest per-axis choices are then consolidated and validated in `res_model`, which additionally carries the single-factor layer-mixer (Section 3.3.1) and encoder (Section 3.4.2) studies that the primitive stacks cannot express cleanly. The same study names are used in the results (Section 4), one table per axis.

DMHAM plays a narrower part: it replicates only the fusion and layer-pooling trends and has no alignment axis. As noted above, its figures are read as directional support for the CogniAlign trends, not a head-to-head comparison.

3.8 Evaluation Metrics and Baselines

The primary metric is classification accuracy, reported in a manner consistent with each cohort's evaluation protocol. For ADReSSo we report both the mean cross-validated validation accuracy and the accuracy on the official `test-dist` partition, the latter being the headline figure comparable with the challenge literature. For the two Spanish tasks, which have no official test set, we report validation accuracy only, computed as the mean of the per-fold best validation accuracy across the five folds, and we never substitute an artificially held-out split for the missing test partition. Alongside accuracy, per-sample validation and test inference times are recorded so that the compute cost of each technique is accounted for rather than assumed; with the frozen encoder outputs precomputed and cached, these times prove comparable across the fusion families, so accuracy remains the deciding criterion (Section 4). The word-to-token alignment of CogniAlign [15] serves as the literature baseline against which the two proposed token-to-token schemes are measured, and challenge-reported systems on ADReSSo provide the external point of comparison for the AD task. Each task is also measured against trivial baselines, the majority class and a uniform-chance predictor, so that the reported accuracies can be read against a non-trivial floor. Confidence intervals and paired significance tests are reported for the headline baselines and contrasts (Section 4.1); elsewhere results are point estimates, with exhaustive per-cell testing left as a limitation revisited in the conclusions (Section 5).

4 Results

This chapter reports the experimental programme laid out in Section 3. It opens by establishing the statistical footing, the trivial baselines each task must clear and the significance of the main contrasts (Section 4.1), then walks the per-axis ablations one design axis at a time, establishing where accuracy comes from across the three tasks (Alzheimer's disease on ADReSSo, amyloid-β status on SPIN, and PPA variant classification on the same SPIN subset), and draws the strongest choices together into a best bundle for each task (Section 4.3). A short study of cached audio augmentation (Section 4.4) closes the chapter.

Two conventions hold throughout. ADReSSo is the only cohort with an official held-out partition, so it is the only task for which a `test-dist` accuracy is quoted alongside cross-validated validation accuracy; the two Spanish tasks are reported on validation folds only, and no artificial test split is ever substituted for the one they lack. Every accuracy is a mean over the five cross-validation folds and the two training seeds (43 and 3407). Confidence intervals and paired significance tests are reported for the headline baselines and contrasts (Section 4.1); they are not computed exhaustively for every cell, a limitation revisited in the conclusions (Section 5).

Unless a table says otherwise, the numbers come from the unified `res_model` implementation run under a single stable recipe: a learnable weighted mixture across the Wav2Vec 2.0 transformer layers, mean pooling over time, and no handcrafted features, so that one axis moves at a time. The modality and feature-integration studies are the deliberate exception, using a neutral multi-head self-attention fusion with the final transformer layer and no alignment so that their comparison turns on the inputs alone, as their captions note. Where the older CogniAlign and DMHAM grids supply something the unified runs cannot, namely a second independent stack for the fusion and pooling trends or the legacy alignment sweep on amyloid, those numbers are brought in explicitly and labelled as such.

4.1 Baselines and significance

Before the per-axis findings, two questions of statistical footing are settled: whether the models clear a trivial baseline, and which axes move accuracy by more than fold-level noise. Table 6 pairs a majority-class and a random-chance baseline for each task with its best validated cell and a 95% confidence interval over the cross-validation folds and seeds. Every task clears its majority baseline by a wide margin, +29.6, +23.5, and +34.2 percentage points on amyloid, PPA, and ADReSSo, each significant at $p < 0.001$ under a paired test over folds.

Paired tests on the two axes the thesis treats as its levers separate them cleanly. Fusion is a real lever: replacing naive concatenation with bidirectional cross-attention lifts accuracy by +17.5 points on amyloid and +18.3 on PPA, both at $p < 0.001$ (Table 9). Alignment is not: the character-proportional token-to-token scheme differs from the word-to-token baseline by no more than about a point on any task, and the difference is not significant ($p = 0.082$, 0.369, and 0.985 on amyloid, PPA, and ADReSSo; Table 11). The subword alignment is therefore best read as an architectural and clarity refinement rather than an accuracy lever, consistent with the small, conditional gains seen in the alignment study. Significance is reported for these headline contrasts and baselines rather than exhaustively for every cell.

Table 6: Trivial baselines and the best validated cell per task (`res_model`). *Majority* is the pooled proportion of the most frequent class over the cross-validation samples; *Chance* is uniform random. The best cell is the mean cross-validation accuracy with a 95% confidence interval over folds and seeds, taken from the stable-recipe bidirectional cross-attention cells used for the paired tests; these are the controlled cells, not the tuned per-task bundles of Table 13 (so amyloid is 81.64% here against the 83.90% tuned run, and ADReSSo is the 86.64% weighted-layer cell rather than the 88.16% sample-mixer one). Δ is the gain over majority; *** marks $p < 0.001$ under a paired test over folds.

Task	Majority	Chance	Best cell (95% CI)	Δ vs majority
Amyloid	52.05	50.00	81.64 [78.8, 84.5]	$+29.6^{***}$
PPA	51.49	33.33	74.95 [70.4, 79.5]	$+23.5^{***}$
ADReSSo	52.41	50.00	86.64 [80.5, 92.8]	$+34.2^{***}$

4.2 Ablation findings

The experiments isolate one design axis at a time, ordered so that each axis is settled before the ones that build on it: first the inputs (which modalities to feed in, whether handcrafted features help, and where those features enter), then how to fuse the streams, then how to pool the Wav2Vec layers and how to align audio to text, and finally which pretrained encoders to use, swapped one at a time within the recipe the earlier axes have settled. Each axis is varied with the others pinned to the stable recipe, so that a change in accuracy can be charged to the axis under test. For the two axes that a second, independent implementation also covers (fusion and Wav2Vec pooling), its agreement with the unified runs is noted. The per-task bundles these findings imply are gathered in Section 4.3.

The modality axis comes first, and it is evaluated under a single fixed fusion so that the comparison isolates the inputs rather than the architecture. Every cell in Table 7 uses the same multi-head self-attention block over the concatenated streams, with the handcrafted features entering as one additional token and the final Wav2Vec 2.0 layer throughout; only the set of modalities changes, so each comparison is single-factor. Because this neutral grid deliberately drops the weighted layer pooling and per-task alignment used in the later studies, its absolute accuracies are not directly comparable with them, so the same multi-head self-attention fusion reads differently in the fusion table (Table 9); the figures here are meant for comparing modalities within the table, not across tables. Read this way, text is the strongest single modality on all three tasks, audio the weakest, and the handcrafted acoustic features on their own are surprisingly competitive, second only to text on amyloid (74.51%). Combining modalities helps, but not monotonically: the best configuration is task-dependent, favouring the full audio–text–features set on amyloid (80.14%), the audio–text pair on ADReSSo (84.01%, with a 76.48% test accuracy), and the text–features pair on the noisy PPA cohort (60.91%). On PPA in particular, adding the weak audio stream on top of text and features does not help in this grid; the apparent ordering among the closest PPA configurations sits within the seed-to-seed noise of that small three-class cohort. The acoustic features' own contribution, isolated under the recommended cross-attention fusion, is the smaller task-dependent effect quantified next in Table 8.

Table 7: Controlled modality ablation per task (`res_model`). Every cell shares one fixed architecture, a multi-head self-attention fusion over the concatenated token streams with the 55-dim handcrafted acoustic features (AF) entering as an additional token, the final Wav2Vec 2.0 layer, and no alignment; only the set of input modalities changes, so each comparison is single-factor. Mean CV validation accuracy (%) over two seeds (43 and 3407). AF's isolated effect under the recommended cross-attention fusion is reported in Table 8.

	SPIN		**ADReSSo**
Input configuration	**Amyloid**	**PPA**	AD
Audio (A)	67.79	54.99	62.00
Text (T)	78.65	59.95	82.50
Acoustic feats (AF)	74.51	57.43	65.30
A + T	79.32	60.69	**84.01**
A + AF	75.24	58.44	64.74
T + AF	79.22	**60.91**	82.80
A + T + AF	**80.14**	56.48	81.89

Text is the strongest single modality on every task; acoustic features alone are competitive, especially on amyloid. The best combination is task-dependent, and on the noisy PPA cohort adding the weak audio stream on top of text and features does not help in this grid. ADReSSo also reports test accuracy for the bimodal cells (Audio+Text 76.48%). AF's controlled effect is in Table 8.

Layered on top of those two streams are the handcrafted acoustic features. The 55 librosa/Praat descriptors are an optional side-channel, and the ablation (Table 8) shows their contribution to be real but task-dependent. Concatenated after pooling, they lift amyloid by 2.1 pp (78.60% to 80.68%) and ADReSSo by 0.6 pp, but cost PPA 2.3 pp. There is no universal gain. How the features are integrated, whether as a pre-fusion token or by post-pooling concatenation, has only a task- and fusion-dependent effect with no consistent winner, so where the features are used at all they are added by the simpler post-pooling concatenation. That is why the recommended default leaves the features off and turns them on only where a task earns it, amyloid being the clear case, consistent with the pause- and voice-quality phenomenology of cognitive decline.

With the inputs settled, the central axis is how to fuse them. Fusion is the axis with the widest spread (Table 9, shown graphically in Figure 11), and the three groups behave exactly as their inductive biases would predict. The two non-sequential baselines and plain self-attention stay near chance on the Spanish tasks; self-attention over an undifferentiated audio–text bag has no way to find the cross-modal structure on its own. Positional information and multiple heads help, but the decisive jump comes from keeping the modalities apart and letting one attend to the other. The cross-attention family (plain, gated, and

Table 8: Handcrafted acoustic-feature (AF) ablation per task (`res_model`, bidirectional cross-attention, weighted Wav2Vec 2.0 layers). 55-dim librosa/Praat descriptors. *None* = no AF; *token* = AF as an extra sequence token; *concat* = AF concatenated after pooling. Mean CV validation accuracy (%) over seeds. Per-task alignment: amyloid word→token, PPA tok→tok (uniform), ADReSSo word→token.

	SPIN		**ADReSSo**
Acoustic features	**Amyloid**	**PPA**	AD
None	78.60	**72.52**	86.94
Token integration	n/a*	n/a*	86.94
Concatenation	**80.68**	70.25	**87.54**

*Token-level AF integration was only swept on ADReSSo. Concatenated AF helps ADReSSo (+0.6 pp) and amyloid (+2.1 pp) but slightly hurts PPA (−2.3 pp): acoustic features are beneficial but task-dependent, not universal.

bidirectional) tops every task: between 81 and 83% on amyloid, the low seventies on PPA, and the high eighties on ADReSSo. Mamba is the telling outlier. It is strong on English and amyloid (85.40% on ADReSSo, 77.49% on amyloid) yet collapses to 52.57% on PPA, and that collapse is what rules it out as a general-purpose default despite its ADReSSo standing.

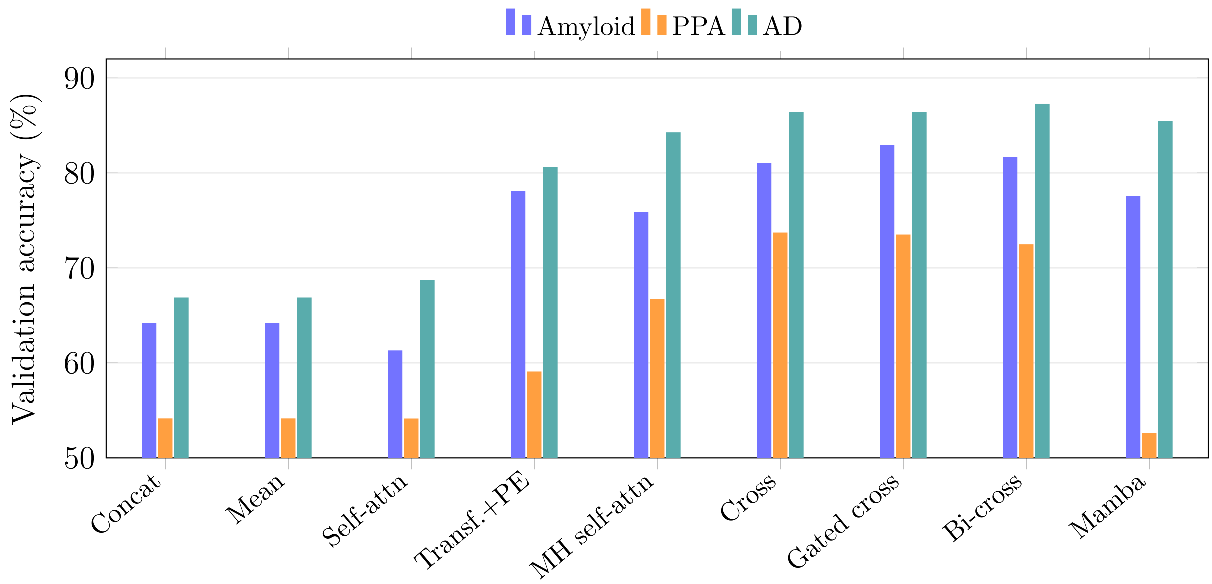

Figure 11: Fusion-family validation accuracy by task (see Table 9).

This ranking is the main trend the second, independent DMHAM stack was built to check, and it survives the change of implementation. In the legacy fusion grid the cross-attention and Mamba families again lead within DMHAM (bidirectional cross-attention reaches 85.21% validation on ADReSSo, for instance), even though DMHAM uses different encoders, a different ASR path, and a shorter training schedule. The absolute numbers diverge between the two stacks, which are not a controlled head-to-head, but the ordering of fusion families holds, and reproducing that ordering twice is the whole point of running it twice. Inference cost is not a discriminating factor among the families: with the encoder outputs precomputed and cached, every fusion variant runs in well under a millisecond per sample, so the choice is driven by accuracy alone.

Table 9: Fusion-technique ablation per task (`res_model`, weighted Wav2Vec 2.0 layers, no acoustic features). The first two rows (concatenation, element-wise mean) are non-sequential baselines; the remaining seven are the sequence-to-sequence fusion families of Table 5. Mean CV validation accuracy (%) over seeds. Fixed per-task alignment: ADReSSo tok→tok (char); amyloid word→token; PPA tok→tok (uniform).

	SPIN		**ADReSSo**
Fusion technique	**Amyloid**	**PPA**	AD
Concatenation	64.12	54.10	66.83
Element-wise mean	64.12	54.10	66.83
Self-attention	61.26	54.09	68.65
Transformer (positional)	78.05	59.04	80.58
Multi-head self-attention	75.85	66.66	84.22
Cross-attention	81.00	**73.67**	86.34
Gated cross-attention	**82.88**	73.46	86.34
Bidirectional cross-attention	81.64	72.43	**87.23**
Mamba	77.49	52.57	85.40

Cross-attention variants dominate on every task; Mamba is competitive on English and amyloid but collapses on PPA. The concatenation and element-wise-mean baselines, and plain self-attention, stay near chance on SPIN.

With a fusion mechanism chosen, attention turns back to the audio branch. Two questions sit on it: whether to read the final Wav2Vec layer or a learnable mixture of all twelve, and, if a mixture, whether a single global set of weights is enough. The pooling table (Table 10) gives a muted answer. A weighted layer stack helps ADReSSo slightly (86.64% against 86.32%), but on the Spanish tasks the final layer is as good or better, clearly so on PPA (74.95% against 72.43%). Depth, once a competent fusion is in place, buys little.

The adaptive layer mixer, evaluated only in the unified implementation, sharpens the picture. Replacing the single global weight vector with a per-utterance *sample* mixer at temperature 0.1 lifts ADReSSo from 87.23% (global mixer, uniform initialisation) to **88.16%**, and the learned weights are interpretable: the mixer concentrates on the top layer (layer 11, weight ≈ 0.20 against the uniform 0.083) while keeping secondary mass on the lowest layer, and it varies those weights from one utterance to the next, with a mean per-utterance weight spread of 0.24, instead of settling on a single profile (Figure 12). The finer *token* mixer does not improve on this, so the per-utterance sample mixer is the variant carried into the recommendation: cheap, input-conditioned, and biased toward the upper layers where lexical content concentrates. This split is what the documented layer specialisation of a self-supervised speech encoder would predict: the lower layers (near layer 0) carry phonetic and speaker detail while the upper layers (near layer 11) carry the more abstract lexical content, so a mixer that puts its primary mass on layer 11 and secondary mass on layer 0 is reading the lexical signal while retaining a low-level acoustic channel, rather than committing to either extreme.

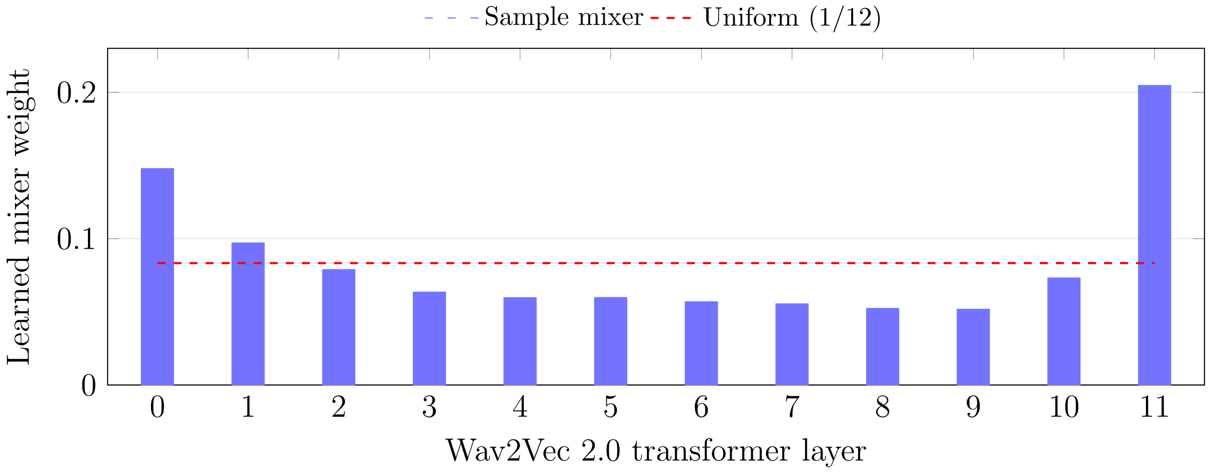

Figure 12: Learned per-layer mixer weights on ADReSSo (seed 43).

Table 10: Wav2Vec 2.0 layer-pooling ablation per task (`res_model`, bidirectional cross-attention). *Final* = last layer; *weighted* = learnable 12-layer mixture. Mean CV validation accuracy (%) over seeds. Per-task alignment: amyloid tok→tok (char), PPA tok→tok (uniform), ADReSSo word→token.

	SPIN		**ADReSSo**
Pooling	**Amyloid**	**PPA**	AD
Final layer	**80.75**	**74.95**	86.32
Weighted layers	80.75	72.43	**86.64**

Weighted layers help ADReSSo slightly; on SPIN the final layer is as good or better, especially on PPA.

The next axis, audio–text alignment, is the thesis's own contribution, so it is examined under two lenses. Under the unified stable recipe (Table 11) the three schemes finish close together on all three tasks: character-proportional token-to-token is best on PPA (73.48%), word-to-token edges amyloid (81.64%), and on ADReSSo the two token-to-token schemes and the baseline are within a few tenths of a point. Subword alignment is therefore not a free lunch under every configuration, and that is worth stating plainly; the difference from the word-to-token baseline is in fact not statistically significant on any task (Section 4.1).

The larger effect appears in the configuration where it should: the legacy alignment grid on amyloid, with handcrafted features and a final-layer audio representation. There, character-proportional alignment reaches 85.72% against 85.13% for word-to-token, with the uniform scheme matching the baseline at 85.13%. The proportional allocation, which gives longer subwords proportionally more of the spoken word's acoustic frames, is the one refinement that helps, and amyloid is where it helps most. Read across both lenses, the honest summary is that token-to-token alignment is a worthwhile, low-cost refinement that pays off on specific task and feature combinations rather than a universal improvement, and that the character-proportional variant is the stronger of the two proposals.

Table 11: Audio–text alignment ablation per task (`res_model`, bidirectional cross-attention, weighted Wav2Vec 2.0 layers). Mean CV validation accuracy (%) over seeds. The token→token schemes are the contribution of this thesis.

		SPIN		**ADReSSo**
Alignment	Source	**Amyloid**	**PPA**	AD
word→token	CogniAlign	**81.64**	72.43	**86.32**
tok→tok (uniform)	This thesis	79.79	72.43	86.02
tok→tok (char)	This thesis	80.75	**73.48**	86.02

Under a stable recipe the three alignments are close on all tasks; character-proportional token→token is best on PPA, while word→token edges amyloid. The larger token→token gain on amyloid appears in the legacy AF + final-layer stack, where the proportional scheme reaches 85.72%.

The last axis swaps the pretrained encoders one at a time (Table 12), and the outcome is firmly task-dependent. ADReSSo is best served by the original lightweight pairing of DistilBERT text with the English Wav2Vec 2.0 base encoder (86.12%; this baseline comes from the encoder sweep and is a different run from the 88.16% adaptive layer-mixer result (Figure 12), so the two are not directly comparable), and degrades when either is replaced, most sharply when the audio encoder switches to the multilingual XLSR-300M (80.37%). The Spanish tasks pull the other way: XLSR-300M audio gives amyloid its best encoder result (81.40%), and ModernBERT text helps PPA (71.70%). The larger cross-lingual audio model earns its place precisely where the language no longer matches the English pretraining, which is the contrast the ablation was designed to expose, and a reminder that the most capable encoder is not automatically the right one once the task is fixed.

Table 12: Feature-extractor (encoder) single-factor ablation per task (`res_model`; fixed recipe: bidirectional cross-attention, word→token, weighted Wav2Vec 2.0 + sample mixer). One component swapped per run; mean CV validation accuracy (%) over seeds 43 and 3407.

			SPIN		**ADReSSo**
Variant	Text encoder	Audio encoder	**Amyloid**	**PPA**	AD
Baseline	DistilBERT	Wav2Vec2 base-960h	79.84	69.94	**86.12**
Audio → XLSR	DistilBERT	Wav2Vec2 XLSR-300M	**81.40**	71.68	80.37
Text → ModernBERT	ModernBERT	Wav2Vec2 base-960h	81.36	**71.70**	77.38
Text → BERT	BERT-base	Wav2Vec2 base-960h	79.88	69.93	81.91

ADReSSo favours DistilBERT + base-960h; on SPIN, XLSR-300M audio (amyloid) and ModernBERT text (PPA) slightly exceed the baseline.

4.3 Results by task

Taken as a whole, the unified `res_model` runs reproduce the main trends established on the two primitive stacks (the cross-attention family wins fusion, depth beyond the final layer adds little, and subword alignment helps narrowly and task-specifically), while adding the two studies the primitive stacks could not express cleanly: the adaptive layer mixer and the single-factor encoder swaps. The grid does not point to one winning configuration but to a different best bundle for each task, collected in Table 13. No two of them share the same recipe, which is the thesis's central empirical message in miniature: the best choice along each axis depends on the task, and a single configuration that wins everywhere does not exist in this grid. What the evidence supports instead is a sensible cross-task default, bidirectional cross-attention, word-to-token alignment, a weighted audio stack with the sample mixer, and handcrafted features off, that stays close to each per-task best while remaining stable across folds, seeds, and tasks. The three bundles are read out below.

Alzheimer's disease (ADReSSo). ADReSSo is the anchor task and the only one with an official held-out partition, so it is the one place a model is judged on a previously unseen split rather than on its own folds. The recommended configuration, bidirectional cross-attention over word-to-token-aligned streams with a weighted Wav2Vec stack refined by the per-utterance sample mixer, reaches **88.16%** validation accuracy, obtained without leaderboard-specific tuning. The official `test-dist` partition tells a complementary story: the Mamba block edges the validation grid (88.44%) and, run through the full-training protocol, reaches **82.39%** on that held-out split (mean over seeds 43 and 3407, 81.69% and 83.10%), among the stronger challenge systems and the only configuration evaluated there. That is a genuine finding; but because the same block collapses on PPA and its ADReSSo lead is seed-sensitive, the recommendation keeps the steadier bidirectional cross-attention.

Amyloid-β status (SPIN). The amyloid task recovers a fluid biomarker from forty seconds of Spanish picture description, with no official test set, so every figure is a five-fold cross-validation accuracy. It leans on the text stream (Table 7), and gated cross-attention is the best plain fusion at 82.88%. Its headline comes from the configuration the thesis contributes: character-proportional token-to-token alignment, a final-layer audio representation, bidirectional cross-attention, and the 55 handcrafted features switched on, reaching **83.90%** (a tuned single run at seed 43, as noted in Table 13, not a two-seed mean). Amyloid is also where the alignment contribution shows most clearly: in the legacy handcrafted-feature and final-layer grid the proportional scheme reaches 85.72% against 85.13% for word-to-token, the largest of the alignment gains.

PPA variant (SPIN). PPA is the hardest task, a three-class problem over a small, imbalanced cohort whose fold-level accuracy is noisy, so its numbers are read as indicative rather than definitive. The best bundle, bidirectional cross-attention with uniform token-to-token alignment and a final-layer audio representation, reaches **74.95%**. Under the controlled modality comparison (Table 7) its two single modalities are close, with text edging audio (59.95% against 54.99%), so no task prefers audio outright and modality leadership for PPA is best read as fusion-dependent rather than intrinsic. PPA is also the only task that punishes the wrong fusion outright, with the Mamba block falling to 52.57%, barely above the majority class.

Table 13: Best technique bundle per task in the unified `res_model` implementation. SPIN (amyloid, PPA) are Spanish, 5-fold CV validation only; ADReSSo (English) additionally reports official test-dist accuracy after full training. Values are means over seeds 43 and 3407, except the amyloid figure, which is a best single run (footnote *a*).

Task	Best technique stack	Val. acc. (%)	Test acc. (%)
Amyloid (SPIN)	bicross, tok→tok (char), final Wv, +AF	**83.90**[a]	n/a
PPA (SPIN)	bicross, tok→tok (uniform), final Wv	**74.95**	n/a
AD (ADReSSo, ref.)	Mamba, word→token, final Wv	88.44	82.39[b]

[a] Best single run (seed 43, lr 1.5×10^{-5}, dropout 0.25) with the 55-d handcrafted acoustic features; the plain-recipe gated cross-attention reaches 82.88%. [b] Official ADReSSo test-dist under the full-train protocol. Mamba tops the ADReSSo grid, but the recommended cross-task default is bidirectional cross-attention with the per-utterance sample mixer (88.16% cross-validated validation accuracy), preferred for its stability across all three tasks rather than its single-task peak.

4.4 Data augmentation

Within the cached augmentation protocol of Section 3.6, augmentation does not improve clean-validation accuracy. Using an augmented view at every training step ($p_{\text{aug}} = 1$) lowers the mean accuracy by 4 to 6 pp on the two SPIN tasks and by about 1 pp on ADReSSo. Mixing clean and augmented views recovers most of that loss: a 30% augmentation share is the best setting found, matching clean accuracy on amyloid (83.85% against 83.90%) without exceeding it, and remaining 1 to 2 pp below clean on PPA and ADReSSo (Table 14). The amyloid sweep (Figure 13) shows the pattern clearly: accuracy stays close to the clean baseline under light mixing and falls sharply once augmentation dominates.

This is consistent with augmentation acting on frozen, pre-computed features rather than on an encoder trained end to end, over small cohorts whose clean baselines are already tuned. The evaluation is on clean validation folds only; whether augmentation improves robustness on noisy or out-of-domain test audio, which is its usual purpose, is untested here and left to future work. The primary results therefore use the clean cached embeddings throughout.

4.5 Error Analysis

A detailed error analysis, including per-class confusion matrices for the three-class PPA task and an inspection of misclassified ADReSSo and amyloid cases, is left as future work. The present study reports the headline contrasts with significance tests (Section 4.1) but no per-class confusion-level breakdowns; characterising systematic failure modes, for example confusion between the logopenic and non-fluent PPA variants or the influence of ASR transcription errors on the text stream, would require the per-sample prediction logs and is identified here as a direction for subsequent analysis rather than reported with fabricated figures.

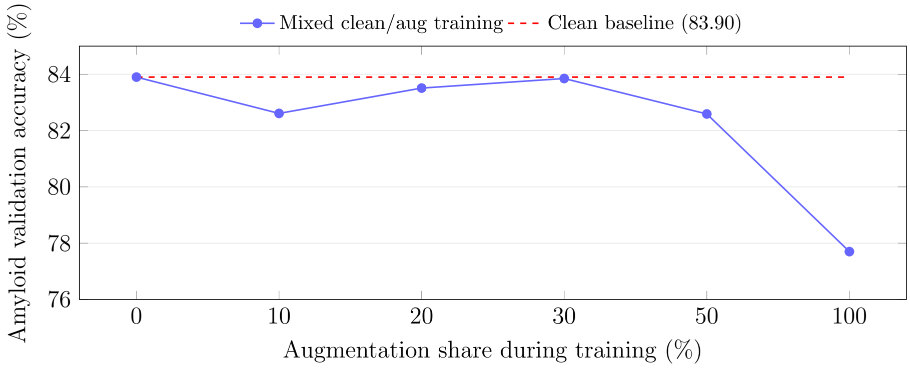

Figure 13: Amyloid validation accuracy versus the training augmentation share.

Table 14: Effect of cached audio augmentation per task (`res_model`, recommended per-task configuration; for ADReSSo this is the bidirectional cross-attention stack at 88.16%, not the Mamba grid leader). Mean validation accuracy (%) over seeds 43 and 3407 (the amyloid clean reference is the seed-43 tuned run). *Clean* = no augmentation; *30% mix* = best augmentation share; *Always* = an augmented view every step.

Setting	**Amyloid**	**PPA**	**ADReSSo**
Clean (no augmentation)	**83.90**	**74.95**	**88.16**
30% mix	83.85	73.70	86.96
Always (100%)	77.70	70.91	87.23

Augmentation never beats clean on validation; light mixing (30%) recovers most of the loss on amyloid but leaves PPA and ADReSSo 1 to 2 pp below clean. Always-on augmentation degrades every task, most on the SPIN cohorts.

5 Conclusions and Future Development

This thesis set out to take a multimodal clinical-speech classifier apart and measure what each of its design choices actually contributes. Instead of reporting a single end-to-end accuracy, it treated the pipeline as a composition of six largely independent axes (modality, fusion, audio–text alignment, Wav2Vec pooling, layer mixing, and feature extractor) and moved them one at a time across three clinical tasks: Alzheimer's disease classification on the English ADReSSo benchmark, and amyloid-β status and PPA variant classification on a Spanish-speaking subset of SPIN. Every technique was studied on two independent stacks and then re-assembled in a single configuration-driven implementation, so that the findings attach to techniques rather than to any one codebase. The objectives set out in Section 1.1 are answered in turn below. The first was a resource contribution, now delivered: the Spanish-speaking SPIN subset, paired with confirmed amyloid-β status and PPA-variant labels and used here for the first time, is curated and characterised in Section 3.2 alongside the public English ADReSSo benchmark.

On the modality axis, text is the strongest single stream on every task and audio the weakest, with the handcrafted acoustic features acting as a task-selective side-channel rather than a universal gain. Across the design axes, the clearest result is that the fusion mechanism dominates every other. Naive combination of the two streams barely clears the better unimodal branch, and plain self-attention over a concatenated audio–text bag stays near chance on the Spanish tasks; the gains arrive only once the modalities are kept apart and made to attend to one another. The cross-attention family leads on all three tasks, and this ordering reproduces under the second, independent stack despite its different encoders and training schedule, which is the strongest evidence in the thesis that the trend belongs to the technique and not to one implementation. The Mamba state-space block is the instructive exception: strong on English and amyloid yet collapsing to chance on PPA, which is exactly why a single-task peak is a poor basis for a default.

Audio–text alignment, the methodological contribution of the thesis, gives a smaller and more conditional gain. Under a stable recipe the word-to-token baseline and the two proposed token-to-token schemes finish within a point of one another on all three tasks, a gap a paired test confirms is not statistically significant, so subword alignment is not a free improvement under every configuration. Where it does help, it helps in the setting it was designed for: in the legacy amyloid grid, with handcrafted features and a final-layer audio representation, the character-proportional scheme reaches 85.72% against 85.13% for word-to-token. The honest reading is that token-to-token alignment is a worthwhile, low-cost refinement that pays off on specific task and feature combinations, and that the character-proportional variant is the better of the two proposals.

The remaining axes move accuracy only at the margin, and always in a task-dependent way. Reading a learnable weighted sum of Wav2Vec layers rather than the final layer changes little once a competent fusion is in place, though the adaptive per-utterance sample mixer adds a small ADReSSo gain (from 87.23% to 88.16%) by concentrating on the upper layers while still varying its weights from one recording to the next. The 55 handcrafted acoustic features help amyloid and, slightly, ADReSSo, but cost PPA accuracy, so they are best left off by default and enabled per task. The pretrained encoders behave the same way: the lightweight DistilBERT and English Wav2Vec 2.0 base pairing is best on ADReSSo, while the cross-lingual XLSR-300M audio encoder helps amyloid and ModernBERT text helps PPA, the larger models earning their place precisely where the language stops matching the English pretraining. Audio augmentation, tested separately under a cached protocol, gave no improvement on clean validation and is left out of the recommended pipeline (Section 4.4).

These observations point to a recommended architecture rather than a single winning run: the configuration specified in Section 3.4.1, built on bidirectional cross-attention, word-to-token alignment, and a sample-mixed weighted audio stack, with the cross-lingual encoder substituted on the Spanish cohorts and the alignment scheme tuned per task where it earns it (character-proportional on amyloid, uniform on PPA, word-to-token on ADReSSo). It reaches 88.16% cross-validated accuracy on ADReSSo, while staying lightweight enough to precompute and cache end to end; the headline 82.39% on the official held-out partition, in line with stronger challenge systems and obtained without leaderboard-specific tuning, belongs to the Mamba block run under the full-training protocol, the only configuration evaluated on that split. Its value is breadth, not a record on any one task: no configuration wins everywhere in this grid, and the central message of the thesis is that the best choice along each axis is task-dependent, with attention-based fusion the one near-universal requirement.

5.1 Limitations

The conclusions should be read against the study's limits. The clinical cohorts are small, and the PPA task in particular is a three-class problem over a few dozen subjects, so its fold-level accuracies are high-variance and are read as indicative rather than definitive. The two Spanish tasks have no official test partition, so they are reported on cross-validation folds only and never against an artificial split. The headline baselines and contrasts are reported with confidence intervals and paired significance tests, but the per-axis grid is not tested exhaustively, so small differences between individual cells should not be over-interpreted. The text stream is derived from automatic speech recognition rather than gold transcripts, which injects a genuine and unmeasured source of error, especially for Spanish and for disfluent aphasic speech. The encoders are frozen throughout, which keeps the comparison clean and the training tractable but also caps the attainable ceiling. Finally, cross-stack figures are directional rather than a controlled head-to-head, and the handcrafted descriptors evaluated here are the 55-dimensional librosa and Praat set, not the standardised OpenSMILE eGeMAPS functionals.

5.2 Future work

Several of these limits map directly onto next steps:

- **Error analysis.** Build the per-class confusion matrices for PPA, where logopenic and non-fluent variants are the likely confusion, and quantify how ASR transcription errors propagate into the text stream and the final prediction.
- **Statistical validation.** Extend the confidence intervals and paired significance

tests beyond the headline baselines and contrasts to every cell of the grid, so that all per-axis differences can be claimed with calibrated confidence.

- **Feature sets.** Re-extract the handcrafted features with OpenSMILE eGeMAPS and compare them against the 55-dimensional librosa/Praat set under identical fusion and alignment settings.
- **Encoders.** Broaden the single-factor encoder sweep to HuBERT and WavLM, and test partial unfreezing or fine-tuning of the acoustic and textual backbones.
- **External validation.** Evaluate on official amyloid and PPA test partitions when available, and on additional cohorts and languages, to confirm that the trends transfer.
- **Regression targets.** Extend the head from categorical labels to a continuous target such as amyloid burden, which the architecture supports by swapping the final layer and loss.
- **Deployment.** Assemble the recommended hybrid into a single end-to-end system with the precompute-and-cache pipeline, as a deployable clinical-screening prototype rather than a research grid.

6 Budget

This is a research thesis built on open-source software and run on a shared university GPU cluster, so its cost is dominated by engineering time, with compute as a secondary item and no licensing expense. Table 15 gives an estimate; the figures are approximate, and cluster usage is valued at a cloud-equivalent rate rather than billed directly.

Table 15: Estimated project budget. Personnel hours span the full thesis period; GPU usage is valued at a cloud-equivalent rate; all software is open-source.

Item	**Basis**	**Cost (EUR)**
Engineering (student)	∼800 h at 25 EUR/h (junior engineer)	20,000
Supervision	∼40 h at 60 EUR/h (senior engineer)	2,400
GPU compute	∼400 GPU-h at 2.5 EUR/h (cloud-equivalent)	1,000
Workstation / laptop	amortised over the project	150
Software	PyTorch, Hugging Face, librosa, Praat (open-source)	0
Total		**23,550**

The frozen-encoder design keeps the compute item small. Because every embedding is precomputed once and cached, the multi-axis ablation grid reuses stored features instead of re-running the encoders, so the GPU budget mostly covers the one-off feature extraction and the training of the lightweight fusion heads rather than the backbones.

7 Environmental Impact

The environmental footprint of this thesis is essentially the electricity used to train and evaluate the models: the data are existing recordings and the software stack is open-source, so no further resources are consumed. Following the usual estimate, the training energy is the product of GPU time, device power, and a data-centre overhead factor (PUE), and the resulting emissions follow from the local grid's carbon intensity.

Taking the roughly 400 GPU-hours of Table 15 at a representative 250 W per GPU and a PUE of 1.5,

$$E \approx 400\,\mathrm{h} \times 0.25\,\mathrm{kW} \times 1.5 \approx 150\,\mathrm{kWh}.$$

The cluster runs on the Spanish grid, whose carbon intensity is on the order of $0.17\,\mathrm{kg\,CO_2/kWh}$, so

$$\text{emissions} \approx 150\,\mathrm{kWh} \times 0.17\,\frac{\mathrm{kg\,CO_2}}{\mathrm{kWh}} \approx 26\,\mathrm{kg\,CO_2e},$$

roughly the emissions of a 200 km drive in an average car.

The figure is deliberately small for a deep-learning project, and the reason is architectural. Freezing the encoders and caching their embeddings means the heavy backbones run once rather than at every training step, so the many-cell ablation grid trains only lightweight fusion heads and adds little marginal energy. The estimate is an order-of-magnitude value: it excludes embodied hardware emissions and development-machine usage, and the grid intensity is an annual average rather than a time-of-use figure.

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Appendices